



Genome-wide cell-free DNA mutational integration enables ultra-sensitive cancer monitoring

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In many areas of oncology, we lack sensitive tools to track low-burden disease. Although cell-free DNA (cfDNA) shows promise in detecting cancer mutations, we found that the combination of low tumor fraction (TF) and limited number of DNA fragments restricts low-disease-burden monitoring through the prevailing deep targeted sequencing paradigm. We reasoned that breadth may supplant depth of sequencing to overcome the barrier of cfDNA abundance. Whole-genome sequencing (WGS) of cfDNA allowed ultra-sensitive detection, capitalizing on the cumulative signal of thousands of somatic mutations observed in solid malignancies, with TF detection sensitivity as low as 10^{-5} . The WGS approach enabled dynamic tumor burden tracking and postoperative residual disease detection, associated with adverse outcome. Thus, we present an orthogonal framework for cfDNA cancer monitoring via genome-wide mutational integration, enabling ultra-sensitive detection, overcoming the limitation of cfDNA abundance and empowering treatment optimization in low-disease-burden oncology care.

Liquid biopsy technology is a transformative force in cancer care¹. Although different analytes are being actively explored for cancer surveillance^{2–7}, detection of somatic mutations in circulating tumor DNA (ctDNA) in plasma cfDNA has been shown to enable noninvasive characterization of the somatic malignant genome^{8–12}. Current applications have often been developed in the context of metastatic high-tumor-burden disease^{13,14}. In this context, cfDNA TF is high, providing for effective mutational characterization with next-generation sequencing (for example, whole-exome sequencing or targeted panels) with minimal modifications¹⁰.

One of the major areas of future promise for cfDNA-based cancer studies is the detection of minimal residual disease (MRD) to guide clinical interventions^{15,16}. However, in the MRD context, cfDNA TF is significantly lower as it is correlated with disease burden¹⁷. To enable mutation detection of low TF cfDNA, the prevailing paradigm has been to increase the depth of sequencing of a limited high-yield target set (for example, common cancer drivers or patient-specific panels sequenced to a depth of tens to hundreds of thousands of reads per base)^{18–20}. Although these state-of-the-art methods provide detection with high accuracy, many patients with radiographically evident disease do not show detectable ctDNA by deep targeted sequencing^{17,18}. These data suggest that ultra-deep sequencing currently underperforms imaging and might be even more challenged in postoperative MRD detection, where substantial debulking likely reduces TF even further. To overcome this

challenge, we present here genome-wide mutational integration for ctDNA detection, whereby breadth supplants depth of sequencing for sensitive detection of low-burden cancer.

Results

Small number of DNA fragments in plasma samples poses a fundamental challenge to deep targeted sequencing in low-burden disease. We hypothesized that deep targeted approaches might be hindered by a fundamental barrier to detection sensitivity—limited input material (the number of cfDNA fragments as measured by genomic equivalents (GEs)). Specifically, in the setting of early cancer or MRD detection, cfDNA abundance might be limiting to deep targeted sequencing, as it imposes a ceiling on effective depth of sequencing (Fig. 1a). To test this hypothesis, we reanalyzed a recent landmark study of early-stage cancer detection using deep targeted sequencing¹⁸. Mutation detection was largely limited to 10^{-3} variant allele fraction (VAF) in cfDNA for all cancer stages (Extended Data Fig. 1a,b, consistent with VAF cutoffs implemented in variant detection¹⁸), despite $\sim 40,000\times$ coverage depth of sequencing and the application of an advanced molecular error suppression²¹. This sensitivity limitation manifested in a stage-dependent decrease in the proportion of patients with detectable ctDNA (Fig. 1b and Extended Data Fig. 1a,b). Consistent with the hypothesis that available input material limits detection, we observed a positive correlation between the number of unique cfDNA reads (representative

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of the number of GEs) in the sample and the number of mutations detected (Fig. 1c), as well as higher unique read coverage in samples with detected ctDNA mutations (Fig. 1d).

To infer a putative underlying TF distribution, we integrated the distribution of the maximal VAF (an approximation of TF) across samples with the fraction of samples that were not detected, with the assumption that detection failures are due to TF below the limit of assay sensitivity (Methods). In advanced disease, higher TF values (median, 4% estimated versus 6% observed in patients with stage IV disease) enabled robust detection in 90% of samples despite the 10^{-3} sensitivity limitation (Fig. 1b, left). However, our analysis suggested that lower TFs in early-stage disease (median TF, 0.02% estimated versus 0.4% observed in patients with stage I disease) result in TF values that are often below the 10^{-3} sensitivity limitation (Fig. 1b, right) and anticipated a substantially increased detection sensitivity through methods capable of detecting TFs as low as 10^{-5} .

These results demonstrated that, in the setting of low-burden disease detection—where TF is low—limited input material likely constitutes a major barrier to the effective application of deep targeted sequencing. As the limited cfDNA input cannot be overcome through sample processing improvement (Methods and Extended Data Fig. 1c,d), we require a rethinking of library and analysis design to maximize the use of available cfDNA fragments (Fig. 1a).

Breadth of sequencing supplants depth of sequencing to overcome the limitation of low number of DNA fragments in plasma samples. Detection of a single somatic single-nucleotide variant (SNV) in a plasma sample results from two consecutive sampling processes, each with its own statistical probability. The first process provides the probability that the mutated fragment is sampled in the limited number of GEs present in a typical plasma sample. The second process is the detection of the mutated fragment in the sample given its abundance, sequencing depth and technical variables, such as sequencing error rate. Although the latter process has been the focus of intensive investigation and technology development (for example, ultra-deep error-free sequencing protocols^{18–20}),

the former stochastic process is infrequently addressed. However, if no physical fragment that supports a targeted SNV is present in the sample, even ideal ultra-deep targeted sequencing will fail to discover a signal. In practice, this problem is further compounded by the fact that a single observation (mutated read) is rarely sufficient for confident detection.

Mathematical modeling of the probability of sampling mutant fragments in a given cfDNA sample (Methods) predicts that the detection probability in TFs relevant to the low-burden cancer regime (TF < 1%, ~5,000 GEs; Fig. 1b,c) will show rapid decline in detection at TFs below 10^{-3} (detection probability as low as 5% for TF = 10^{-5} ; Fig. 1e, left). We note that these limitations are observed even under idealized conditions of (1) exhaustive sequencing; (2) efficient use of all GEs, assuming perfect recovery; and (3) detection based on a single supporting DNA fragment with no sequencing errors. These results show that the plasma sampling process imposes a formidable barrier to mutation detection at low TF regimes, such as in residual disease.

Conversely, our model also shows that the sampling limitation can be effectively overcome by increasing the number of detected sites (SNVs) through increased breadth. Although the detection probability of a large targeted panel (1 Mbp of total coverage and patient mutation load of 10 Mbp, corresponding to 10 SNVs), still showed rapid decline in detection at TFs below 10^{-3} (detection probability lower than 40% for TF = 10^{-5} ; Fig. 1e, middle), integrating over 10,000 SNVs (mutation load of >4 Mbp found in over a quarter of human cancer²²) can provide high detection probability even at TFs as low as 10^{-5} , at a modest sequencing effort readily achieved with standard WGS (detection probability of >85% with 20× depth of coverage or >99% with 50×; Fig. 1e, right).

Development and validation of ultra-sensitive ctDNA detection via genome-wide mutational integration (MRDetect). Our model predicts the TF as a function of number of detected sites, mutation load and sequencing coverage depth. To validate this model, we simulated ctDNA detection using *in silico* admixtures of tumor and normal WGS data from eight patients with various cancers (Supplementary Table 1), generating over 700 *in silico* admixtures

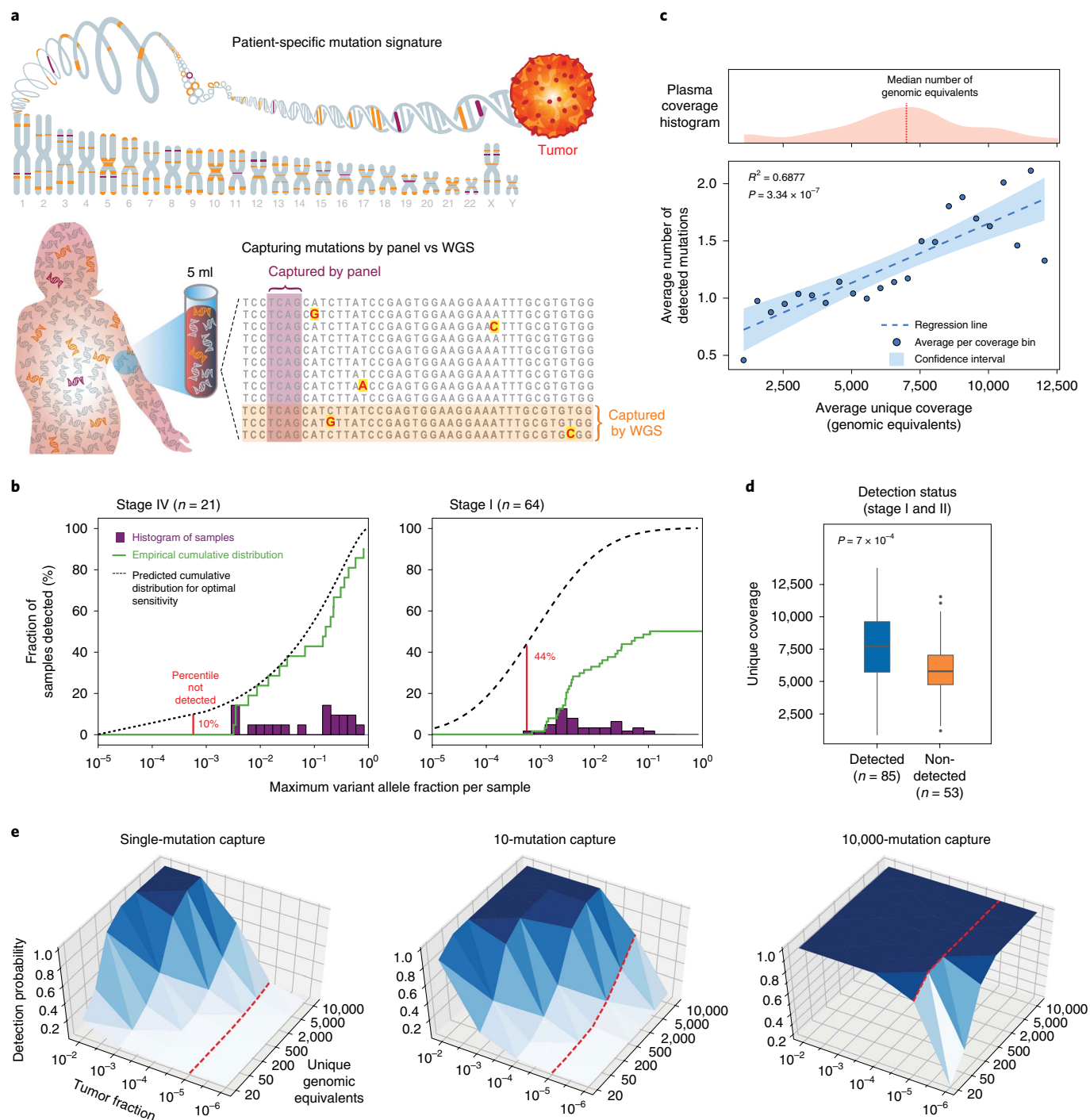
Fig. 1 | Low cfDNA input material limits sensitive ctDNA mutation detection with deep targeted sequencing but can be overcome by genome-wide mutational integration. **a**, Illustrative schematic showing that mutations captured by targeted panel deep sequencing are only a small fraction of genome-wide somatic tumor mutations (upper panel). The low input and random sampling nature of cfDNA collection by blood samples results in higher probability for mutation capture by WGS than by targeted panels (lower panel). **b–d** constitute a reanalysis of previously published data¹⁸. **b**, Histogram of samples detected as a function of the maximum VAF in stage IV (left) and stage I (right) cancer. Maximum VAF serves as an estimation of TF. Patient histogram (solid bars), empirical cumulative distribution (solid line) and predicted ‘optimal-sensitivity’ detection (dashed line; see Methods) are shown. The lower limit of detection (red solid line) shows a residual of 44% of patients with stage I disease that is not detected by the deep targeted sequencing panel (10% for stage IV). Most of these cancers are predicted to have been detected if the VAF lower limit of detection was 10^{-5} . Reanalysis across 64 stage I and 21 stage IV patient samples across tumor types. **c**, Plasma input material (depth of coverage) correlates to the number of detected tumor mutations (correlation and statistical significance calculated with a two-sided Pearson’s correlation). Upper panel, distribution (histogram) of unique GEs (unique coverage, x axis) across all patient samples ($n = 194$ patients, across different tumor types and disease stages) sequenced with exhaustive deep sequencing (median, 40,729×). A median of 6,000 (range, 400–14,000) unique reads (GEs) was found in a median of 4.2 ml of plasma (range, 1.6–9.8). Lower panel, the average number of mutations detected per patient shows a positive correlation with the number of unique GEs in the patient plasma, indicating that input material is a potential limiting factor for mutation detection with deep targeted sequencing (two-sided Pearson’s correlation). The dashed line shows the regression line, and the shaded area represents the 95% confidence interval. **d**, Patient samples with a higher number of unique GEs (coverage) have higher probability for cancer detection by deep targeted sequencing ($P = 7 \times 10^{-4}$, two-sided Wilcoxon rank-sum test). Analysis includes all detected ($n = 85$) and nondetected ($n = 53$) patients with stage I and II disease across tumor types. Box plots represent median, bottom and upper quartiles; whiskers correspond to $1.5 \times$ IQR. **e**, Predicted probability (analytic derivation from binomial distribution of Bernoulli trials; see Methods) of detecting at least one mutated (SNV supporting) read as a function of the number of unique GEs (coverage) and the fraction of ctDNA in cfDNA (TF). Left, targeting only a single SNV (for example, with digital droplet PCR), the probability for detection decreases rapidly with tumor fraction (for example, probability to detect at 10^{-5} with 5,000 GEs is <0.05, red line). Middle, targeting ten patient-specific mutations (for example, 1-Mbp panel for patients with 10-Mbp mutation load or a typical exome-based, patient-specific panel¹⁷) shows low probability of detection in low TF (for example, probability to detect at 10^{-5} with 5,000 GE is <0.4, red line). Right, targeting 10,000 patient-specific mutations (for example, WGS for patients with >4-Mbp mutation load) shows significant probability of detection in low TF even with modest coverage (for example, probability to detect at 10^{-5} is >0.99 with 50× coverage, red line). IQR, interquartile range.

varying the TF mixtures (range, 10^{-5} –0.2) in 35× coverage with replicates ($n = 11$; Methods, Fig. 2a and Supplementary Table 1).

We designed a tumor-informed detection approach for the MRD setting—MRDetect—that uses the primary tumor mutational compendium as a prior. In silico plasma mixtures were then evaluated by searching all sites from the matched patient tumor compendium for corresponding mutations. Because integrating many sites also results in accumulation of sequencing errors, we estimated the extent of noise in WGS-based cfDNA detection in a complementary data set from matched germline WGS (TF = 0, varying sequencing depth (5–35×), in subsampling replicates ($n = 20$), $n = 341$ in total). Signal to noise was measured by comparing the rate of detection (sites with a matching variant out of all sites in the patient-specific

compendium) in plasma admixtures (TF > 0) compared to controls (TF = 0), with detection of TFs as low as 10^{-3} even with only 35× depth of coverage WGS (Fig. 2b).

Further increase of the sensitivity of genome-wide SNV detection required reduction of the noise caused by sequencing errors (~1 per 1,000 bases²¹; Fig. 2c). Lower sequencing quality metrics are associated with sequencing errors, suggesting the possibility to optimize the tradeoff between the true-positive and false-positive rates. (Extended Data Fig. 2a–c). To integrate multiple quality metrics to inform the probability of error in a single read, we implemented a support vector machine (SVM) classification framework (Methods and Extended Data Fig. 2d). We note that this method was specifically designed to address the challenging scenario where the TF is far lower



than the inverse of the depth of coverage, such that, at best, one supporting read is expected for each detected somatic mutation (hence, read-classification rather than the standard locus-classification frameworks^{23,24}). Applying this filtering strategy further reduced the sequencing noise by a median of 14.4-fold (range, 11–17) to lower than 1 per 10,000 bases (Extended Data Fig. 2e).

Notably, this machine learning approach improved the sensitivity of the MRDetect-SNV down to TF of 10^{-5} (Fig. 2b and Extended Data Fig. 2f–h), with assay-level specificity of 95% as evaluated across 160 control WGS. As anticipated, this read-centric method showed superior performance compared to a leading locus-centric method in detecting mutations in this unique context of ultra-low TF samples (Extended Data Fig. 2i). We further note the high concordance between the known synthetic TF admixtures and our model TF predictions (Fig. 2d,e and Extended Data Figs. 2j and 3a–h). Thus, MRDetect also offers TF inference that is not limited by the number of GEs and is therefore orthogonal to current prevailing methods that use mutational VAF.

As detection sensitivity using genome-wide SNV integration is a function of the mutation load and the sequencing depth, we systematically evaluated the relationship between TF's lowest limit of detection (LLOD) and these parameters. Using synthetic admixtures of a melanoma tumor/normal pair to generate plasma-like samples with varying depth of coverage (10–120×), mutation load (2,000–63,000 through subsampling of the mutations detected in the tumor) and TFs (10^{-6} – 10^{-3}), we demonstrated that, at high mutational load, TF detection as low as 10^{-6} is feasible with 120× sequencing depth (Fig. 2f). Collectively, these results suggest that genome-wide mutational integration offers an orthogonal route to ultra-sensitive and quantitative TF monitoring in the low-disease-burden setting.

Expanding the genome-wide mutational integration approach to copy number alterations. Substantial aneuploidy is observed in 88% of human cancer²⁵ (Extended Data Fig. 4), through which large swaths of the genome undergo amplification and deletion, yielding a potentially robust signal for ctDNA detection. We therefore developed MRDetect-copy number alteration (CNA) to integrate changes in read depth at patient-specific amplification and deletion segments across the genome. At these CNA segments, the coverage

in the plasma represented a mixture of the diploid normal cfDNA, together with ctDNA with directional coverage skews (that is, greater read depth in amplified regions and lower read depth in deleted regions). Even at low TF admixtures, we observed sparse read depth skews that were biased toward the known patient-specific CNA segment directionality (that is, amplification or deletion; Fig. 3a). We therefore integrated read depth skews in tumor-specific CNAs across the genome to achieve enhanced ctDNA detection.

We observed high concordance between input synthetic admixture TF and TF inferred by MRDetect-CNA (Fig. 3b and Extended Data Fig. 5a–c), whereas tumor copy-neutral regions, where no directional coverage skews are expected, showed no change in detection signal in relation to TF (Fig. 3c and Extended Data Fig. 5d–f). A high concordance was also noted between the orthogonal SNV- and CNA-based estimates (Extended Data Fig. 6a). CNA-based detection thus enabled sensitive detection at TF as low as 5×10^{-5} (range, 10^{-5} – 10^{-4} ; Fig. 3b and Extended Data Fig. 5a–c), extending the detection range by two orders of magnitude compared to a leading CNA ctDNA algorithm¹⁰ (Extended Data Fig. 6b). To systematically evaluate the relationship between LLOD and the footprint of the aggregated CNAs, we downsampled the cumulative size of segments affected by CNA used for detection. This analysis shows the expected relationship whereby a larger cumulative size of genome affected by CNAs translates into greater sensitivity, with highest sensitivity (TF < 5×10^{-5}) with 1-Gbp CNA footprint (Fig. 3d), applicable to a substantial proportion of patients across tumor types (Extended Data Fig. 4).

Application of the genome-wide mutational integration approach to clinical plasma samples. To evaluate MRDetect in the clinical setting, we tested the method on clinical cohorts comprising patients with lung adenocarcinoma (LUAD; $n=39$), colorectal cancer (CRC; $n=19$) and melanoma ($n=2$) undergoing surgery (patients with LUAD and CRC) or immunotherapy (patients with melanoma). We also collected plasma from controls with no known malignancy ($n=38$), with similar age and tobacco exposure, to characterize the noise background and estimate clinical specificity (Supplementary Tables 2 and 3).

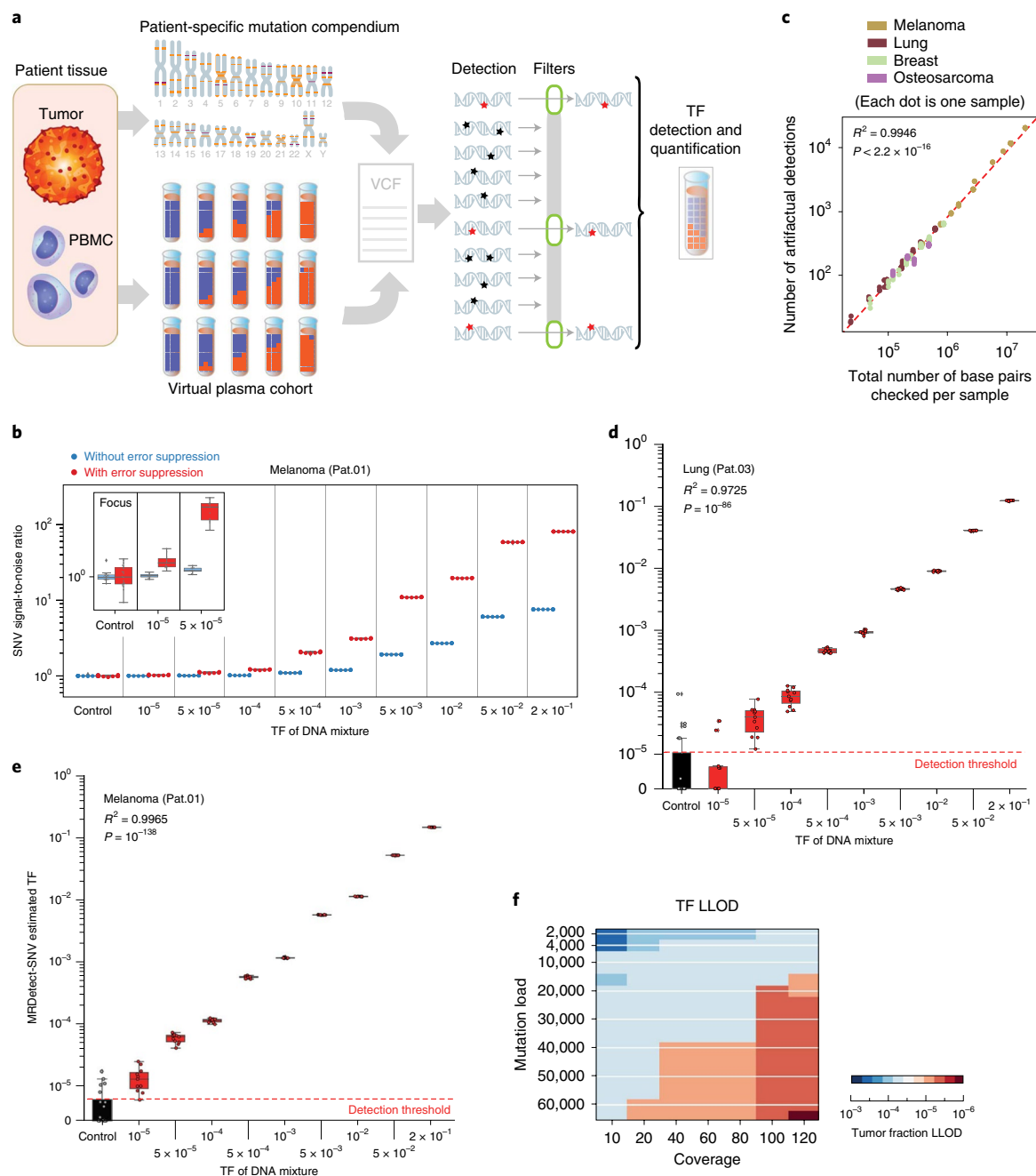
Sequencing noise in cfDNA WGS was similar to noise rates obtained from the *in silico* DNA mixtures (median error rate,

Fig. 2 | Patient-specific genome-wide SNV integration provides ultra-sensitive ctDNA detection and precision TF estimation. **a**, Illustration of genome-wide SNV integration for inference of plasma TF and analytical validation scheme. Patient-specific SNV mutations are detected from matched tumor and PBMC germline WGS. Tumor somatic SNVs are then used to calculate the genome-wide tumor signal in the patient's plasma sample. To test this approach, for each cancer, tumor and PBMC WGS reads were mixed to generate a range of TFs (10^{-5} –0.2) in 35× coverage with multiple replicates ($n=11$; Methods). Across eight patient samples with four different tumor types (lung, melanoma, breast and osteosarcoma), we generated over 700 *in silico* admixture samples. Detection and filtering were applied on each patient sample to benchmark the TF detection sensitivity and accuracy. **b**, Patient-specific SNV signal-to-noise quantification over a range of TFs (10^{-5} –0.2) compared to basal noise signal detected in control (TF = 0) samples (left column), estimated using melanoma sample (Pat.01). Signal to noise was estimated by calculating the log difference between the number of detections in each plasma sample (TF > 0) and the mean number of detections in the controls (TF = 0). Inset panel shows discrimination of tumor and control samples down to TF 10^{-5} after using machine-learning-based sequencing error suppression (red) versus reduced sensitivity with the raw unfiltered data (blue). For TFs > 0, $n=11$ independent admixture samples. For the control (TF = 0), $n=20$ independently downsampled PBMC replicates. **c**, Correlation between the number of base pairs evaluated for SNVs and the number of mismatches detected (artifactual detections), measured over all synthetic control plasma (no tumor DNA, TF = 0, $n=341$) from eight patients and mutational compendia from four tumor types (lung, breast, melanoma and osteosarcoma). Results show a constant error rate, independent of tumor type, corresponding to previously published^{21,40} Illumina sequencing error rate estimates (~1 per 1,000 bps) (correlation and statistical significance calculated with a two-sided Pearson's correlation). **d–e**, TF inference using genome-wide SNV integration for lung cancer (Pat.03) (**d**) and melanoma (Pat.01) (**e**) samples shows accurate TF estimation as low as 5×10^{-5} and 10^{-5} , respectively, discriminated from control (TF = 0) samples (left box plot). High Pearson's correlation (two-sided test) between the input TF mixture (x axis) and the SNV-based estimated TF prediction confirms accurate inference based on genome-wide mutational integration. For TFs > 0, $n=11$ independent admixture samples. For the control (TF = 0), $n=20$ independently downsampled PBMC replicates. **f**, The LLOD was empirically measured as a function of the input mutational load and the WGS coverage, showing that, for high mutational loads (~60,000 mutations), achieving sensitivity nearing 10^{-6} is feasible. Analysis was done over 21,420 *in silico* admixtures, varying TF (10^{-6} – 10^{-3}), coverage (10–120×) and mutation load (2,000–63,000) and across 20 replicates (random read downsampling) for each admixture. LLOD was defined by the lowest TF that showed significant z-score separation from the control (TF = 0) cohort, for the same context (mutation load and coverage depth; Methods). Throughout the figure, box plots represent median, bottom and upper quartiles; whiskers correspond to 1.5× IQR. IQR, interquartile range.

1.03×10^{-3} (1 bp)); Fig. 4a, unfiltered), allowing for facile adaptation of our SVM de-noising strategy using a training set of control plasma samples ($n=8$), with similar suppression performance across test control samples ($n=30$; Fig. 4a). We further leveraged the short size of cfDNA fragments (~ 165 bp) to detect discordance between read pairs (150-bp paired-end protocol) indicative of potential sequencing errors (Extended Data Fig. 6c). This combined filtering scheme showed a reduction of the sequencing error rate to a median of 4.96×10^{-5} (median, 21-fold reduction; Fig. 4a, filtered), with analytic specificity (probability of a true-negative detection per locus sequenced at $30\times$) of 99.85%. The application of MRDetect-CNA to this cohort required additional adaptations (Methods) to account for differences in read depth profiles in WGS cfDNA compared to WGS from genomic DNA (Extended Data Fig. 6d,e), possibly due to differences in library preparation (PCR versus PCR-free protocols) and

nonuniformity of cfDNA coverage driven by DNA degradation and epigenetic features (for example, chromatin accessibility)⁶.

We first applied MRDetect to the challenge of monitoring immunotherapy response in high-disease-burden metastatic melanoma. Patient-specific mutational compendia were obtained from baseline (pre-treatment) plasma WGS and applied to plasma samples during therapy. The z score between the patient plasma signal and the test control plasma noise distribution ($n=30$) was used to measure the significance of discrimination between ctDNA detection in patient versus control samples (Fig. 4b and Extended Data Fig. 7). These data demonstrated that MRDetect effectively tracked tumor responses, matching radiographic changes, but with higher temporal resolution. Notably, whereas disease burden estimated by MRDetect decreased with therapy, our data supported residual disease detection, consistent with imaging results and in contrast to



benchmarked methods where ctDNA burden decreased below the limits of detection (Fig. 4b, lower panel).

Genome-wide mutational integration enables detection of postoperative residual disease. One of the foremost challenges for residual disease monitoring is detection shortly after surgery to aid with the application of adjuvant therapy in patients with a high risk of relapse (Fig. 4c). In this clinical scenario, resected tumors are sequenced to define the patient tumor-specific mutational profiles, which can then be used to detect minute amounts of ctDNA in the patient plasma postoperatively.

To evaluate MRDetect in the postoperative MRD setting, we tested CRC cfDNA, where postoperative ctDNA detection was shown to predict adverse outcome²⁶. We collected plasma from a cohort of 19 operable patients with CRC before surgery and at a median of 43 d after surgery (Supplementary Tables 2 and 3), including six CRCs with microsatellite instability and 13 microsatellite stable tumors (Fig. 4d and Supplementary Table 4). To further ensure the robustness of the ctDNA detection, we bootstrapped the patient plasma sample by downsampling the WGS data (down to 80% coverage, $n=20$ random downsampling replicates), confirming that detection is not driven by isolated sites (Extended Data Fig. 8a). MRDetect-SNV detection performance was evaluated using a receiver operating curve (ROC) analysis showing area under the curve (AUC) $\pm$ s.e. = 0.97 ± 0.025 (sensitivity $\pm$ s.e. = $90\% \pm 0.069\%$, specificity $\pm$ s.e. = $98\% \pm 0.006\%$; Extended Data Fig. 8b). An orthogonal noise model can be generated by applying the patient-specific mutational compendium to other plasma samples of patients with CRC, which might reveal artifactual non-patient-specific detection due to matching mutational sequence context preference (for example, microsatellite instability signature). This cross-patient analysis showed similar detection profiles and performance (Extended Data Fig. 8c,d), supporting detection that is highly specific to an individual malignancy.

MRDetect-CNA showed lower performance (AUC = 0.73 ± 0.064 , sensitivity = $40\% \pm 0.126\%$, specificity = $92\% \pm 0.013\%$; Extended Data Fig. 8e,f), consistent with the lower CNA load in CRC (Extended Data Fig. 4f). However, as the MRDetect-SNV and -CNA signals are independent, they can be combined to a single statistical detection score (Methods), showing robust performance in operable CRC detection (Fig. 4e).

After the validation of sensitive and specific preoperative plasma detection, we tested the more challenging task of detecting ctDNA postoperatively. Postoperative detection was evaluated in the same CRC cohort ($n=19$; Fig. 4f and Extended Data Fig. 8g,h) and was found to be associated with shorter disease-free survival (Fig. 4g; see

Supplementary Table 5 for analysis accounting for additional clinical features) over a median postoperative follow-up of 15 months (range, 4–32; Fig. 4g). Recurrence was not observed in any of the 12 patients in whom ctDNA was not detected in postoperative plasma. Four patients with postoperative detection did not show evidence of recurrence, of whom three received adjuvant therapy that may have eliminated residual disease, consistent with prior reports showing that adjuvant therapy reduces recurrence even in patients with detectable postoperative ctDNA²⁶. The remaining patient had a microsatellite instable tumor that was shown to be associated with extended course to relapse²⁷ and even rarely with spontaneous regression^{28,29}. Alternatively, this patient might reflect a false-positive classification.

In addition, we tested ctDNA detection in the challenging case of low-burden-disease LUAD, where even tumor-informed targeted panel detection showed limited preoperative and postoperative sensitivity¹⁷. We collected plasma from a cohort of patients with early-stage LUAD (78% stage I–IIa LUAD) before surgery and at a median of 2.5 weeks after surgery (Fig. 5a and Supplementary Tables 2 and 3). MRDetect-SNV applied to preoperative plasma (Fig. 5b, Extended Data Fig. 9a and Supplementary Table 4) showed AUC = 0.82 ± 0.049 (Extended Data Fig. 9b). TF estimates using our mathematical model showed the expected correlation between ctDNA TF and disease stage (Fig. 5c), as well as confirmed that ctDNA TF in early-stage LUAD can be often $<10^{-3}$, in agreement with the above-described modeling result (Fig. 1b,c). Performance was maintained in the cross-patient analysis (Extended Data Fig. 9c,d), validating MRDetect high specificity even in a more challenging setting. MRDetect-CNA showed similar performance to CRC (Fig. 5b and Extended Data Fig. 9e,f), with integration of SNV and CNA detection leading to improved performance in early-stage LUAD detection (Fig. 5d). Increasing the detection threshold to achieve higher clinical specificity of $96\% \pm 0.006\%$ still maintained high LUAD detection sensitivity of $67\% \pm 0.079\%$ (Fig. 5d). These data show that genome-wide integration for disease monitoring can perform as well or better than deep targeted approaches^{17,18} or other CNA-based methods¹⁰ (Extended Data Fig. 9g), while overcoming the limitation of input GE abundance.

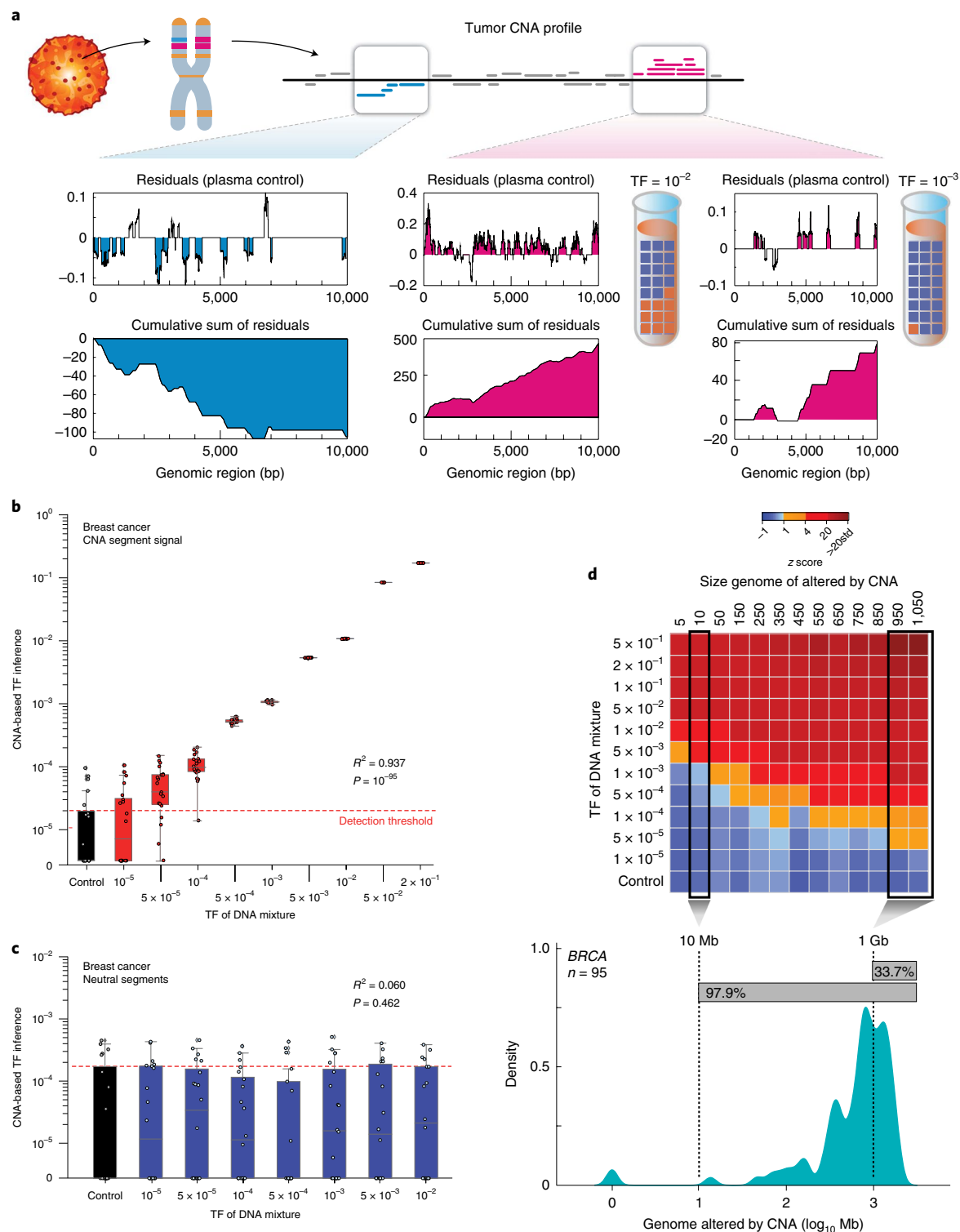
Postoperative detection was evaluated for a cohort of early-stage postoperative samples ($n=22$; note that these patients were from the same cohort evaluated in the preoperative detection, except three patients (LUAD37–39) for whom a preoperative sample was not available). ctDNA was detected in ten postoperative plasma samples by MRDetect (SNV or CNA method; Fig. 5e and Extended Data Fig. 9h,i), associated with shorter disease-free survival (Fig. 5e,f; see Supplementary Table 5 for analysis accounting for

Fig. 3 | Patient-specific genome-wide CNA integration provides ultra-sensitive ctDNA detection and precision TF estimation. **a**, Illustration of the genome-wide CNA integration for the inference of plasma TF. Patient-specific CNA segments are detected from matched tumor and germline WGS. Tumor CNA segments are then used to calculate the genome-wide accumulated signal in the patient's plasma sample. Amplification segments (right panels, pink) in synthetic plasma with TF of 1% and 0.1% show sparse TF-correlated signal that can be integrated owing to the positive read coverage bias. Deletion segments (left panel, blue) show similar sparse signal, which is negatively biased. Upper panel, normalized coverage difference between plasma and control (TF = 0) across a 10-Kbp region. Lower panel, sum of coverage differences between plasma and control. **b**, Using patient-specific CNA mixture model allows for the integrated CNA signal to be directly converted to TF estimates in a breast cancer (Pat.05) in silico synthetic plasma sample, with accurate TF estimation as low as 5×10^{-5} , discriminated from the control samples (TF = 0, left box plot). High Pearson's correlation (two-sided test) between the input TF mixture (x axis) and the CNA-based estimated TF (y axis) shows accurate inference based on CNA patterns. $n=20$ for each TF > 0 sample, as well as the control (TF = 0) samples. **c**, Using the CNA mixture model on copy-neutral regions of the same tumor genome (that is, regions that do not show differential read coverage between tumor and germline WGS), do not show correlation with TF (two-sided Pearson's correlation). $n=20$ for each TF > 0 sample, as well as the control (TF = 0) samples. **d**, CNA method LLOD was empirically measured as a function of the CNA load showing high sensitivity (5×10^{-5}) for tumors with a copy number variation footprint of 1 Gb or more. Upper panel, ctDNA detection at various TF (10^{-5} – 5×10^{-1} and control (TF = 0)) and CNA load (5–1,050 Mb) estimated in a breast cancer (Pat.05) in silico synthetic plasma samples. Detection is observed at TF = 5×10^{-3} at 5 Mb and as low as TF = 5×10^{-5} at 1 Gb (z score ≥ 1). Lower panel, smoothed histogram (kernel density estimate) of a representative cohort of WGS breast cancer (BRCA) samples from The Cancer Genome Atlas ($n=95$). Most (97.9%) of the samples have CNA load above 10 Mb, and 33.7% have CNA load above 1 Gb, corresponding with ultra-high ctDNA sensitivity. Throughout the figure, box plots represent median, bottom and upper quartiles; whiskers correspond to $1.5 \times$ IQR. IQR, interquartile range.

additional clinical features). Thus, MRD ctDNA detection anticipated recurrence shortly postoperatively, compared to delayed identification with imaging (Extended Data Fig. 10a–d). The cohort of patients with ctDNA detected postoperatively also included five patients with no evidence of recurrence, including a patient who was treated with adjuvant chemoradiation after plasma collection. The additional four patients with no evidence of recurrence might reflect shorter follow-up time than needed to demonstrate recurrence in these early-stage malignancies (for example, LUAD11 with

only 12 months of follow-up, below the median time for recurrence³⁰), with longer follow-up potentially revealing that residual disease detected anticipates additional recurrences or, alternatively, might reflect false-positive testing. In contrast, the 12 patients in whom ctDNA was not detectable in postoperative plasma showed no recurrence over a median follow-up of 18 months postoperatively (range, 6–36; Fig. 5e).

Notably, it has been shown that ctDNA fragments are shorter than other cfDNA fragments⁶, and thus this feature may even



be used for detection of high-burden ctDNA³⁰. Consistent with this notion, tumor-specific mutations detected in patient plasma showed a shorter fragment size compared to non-patient-specific (cross-patient) mutations that are detected in the same samples (Extended Data Fig. 10e–h). We further developed a kernel density estimator (KDE) trained on patient-derived xenograft (PDX) samples to discriminate between tumor-derived and normal-derived cfDNA based on the fragment size signature (Methods and Extended Data Fig. 10i). Applying this KDE method to the detected preoperative and postoperative samples showed significant shift toward tumor fragment size signature in the tumor-specific mutations versus non-patient-specific (cross-patient) mutations (Fig. 5g and Extended Data Fig. 10j), providing an orthogonal confirmation that the detected mutations are indeed ctDNA derived and further suggesting the potential of future integration of fragment length into machine learning de-noising approaches.

Discussion

Deep targeted sequencing approaches are limited in the context of low disease burden owing to (1) low fraction of ctDNA, often below 0.1%, and (2) limited numbers of available DNA fragments in a typical plasma sample, often in the range of only hundreds to several thousand GEs. The limiting number of GEs effectively places a ceiling on the depth of sequencing beyond which available distinct fragments are exhausted. Thus, in low-disease-burden cancer, deep targeted sequencing is limited to ctDNA detection in frequency above 10^{-3} , resulting in detection in only 20–70% of cases with radiographically demonstrated early-stage cancer^{17–19}. Residual disease (for example, postoperative) detection is anticipated to pose an even greater challenge to current technologies²⁶. To overcome the fundamental limitation of low input of cfDNA, we integrated genome-wide mutational signal to allow ultra-sensitive detection as well as quantitative dynamic monitoring of disease burden. Our results show that, through genome-wide mutational integration, we enabled accurate and sensitive ctDNA detection in fractions as low 10^{-5} . As this was achieved with a modest sequencing depth (35×), the genome-wide approach severs the limiting tie between available GEs and detection sensitivity.

WGS also enables effective integration across orthogonal data dimensions such as SNV and CNA, allowing clinical application to a wide range of tumor types that have either high mutation load²² or

aneuploidy²⁵. WGS provides an abundance of additional information sources to further increase sensitivity, including shorter fragment length of ctDNA compared to normal cfDNA and nucleosome position information^{6,27,29,31,32}. All these data dimensions could be potentially extracted from the same WGS data, allowing for versatile and cost-effective use of a simple experimental workflow.

We note that, although our approach provides sensitive detection of ctDNA, it provides limited confidence in the sensitivity to detect any individual site (for example, a driver mutational event). This suggests that, in scenarios where therapeutic targeting requires such information, deep targeted approaches are more appropriate. However, in the context of the broad cancer detection challenge, this might serve as a potential advantage. Genome-wide detection is often based on the majority of events that are ancestral (that is, preceding transformation^{30,32}) and thus less affected by clonal diversification, which might hinder detection via personalized targeted panels due to spatial subsampling or clonal tides. Moreover, targeted cancer gene panel mutation detection may be confounded by clonal growths of hematopoietic origin^{18,33,34}. This finding might present a larger limitation to panel strategies, given the emerging challenge to distinguish pervasive clonal outgrowths across tissues in physiological aging^{35–37} from true malignancies. In contrast, MRDetect leads to exquisite specificity driven by the patient-specific mutational compendia, such that, even when applied cross patient to samples from individuals with a known malignancy, specificity is maintained.

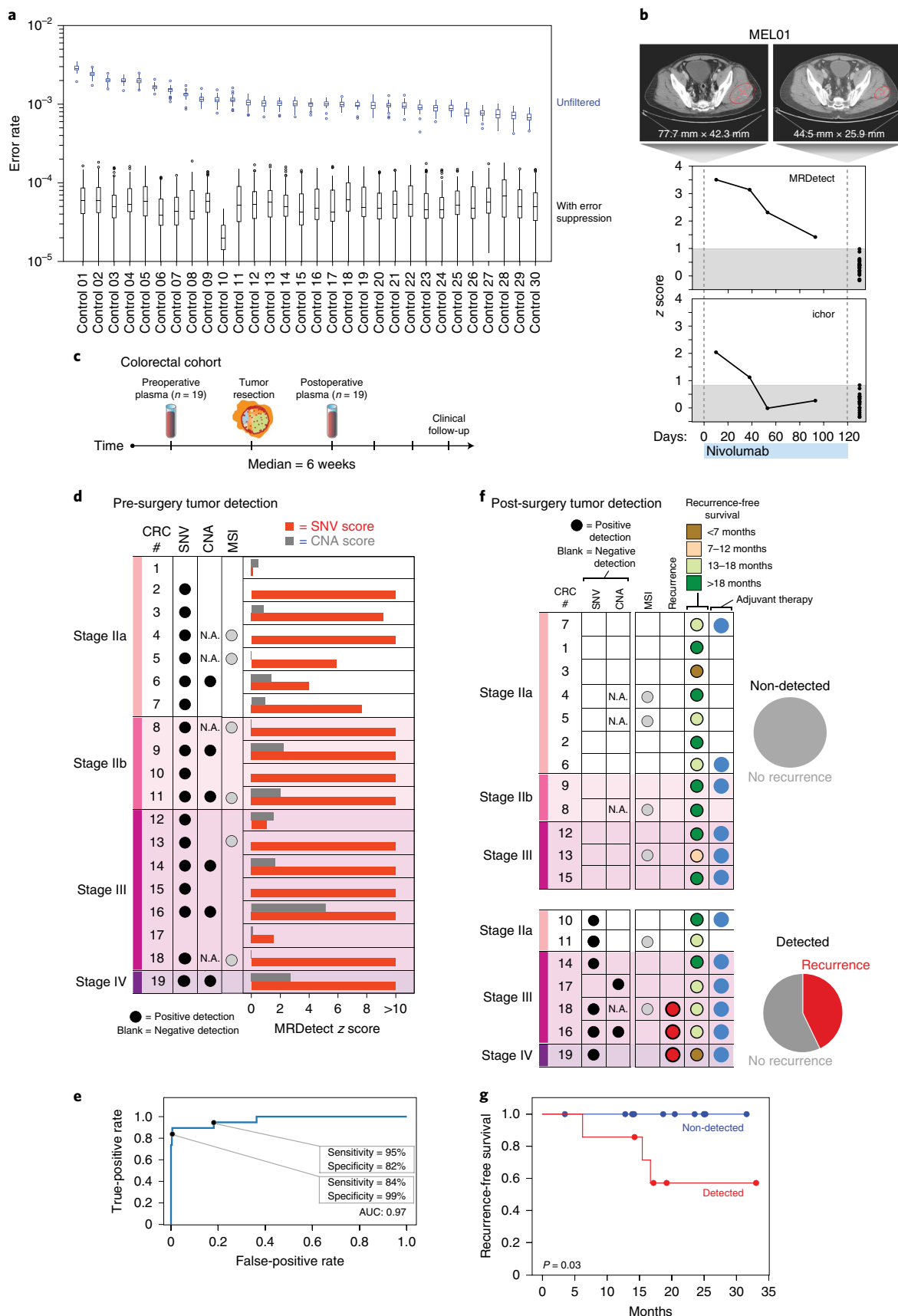
MRDetect further obviates the need for optimizing the design of capture panels and molecular de-noising, enabling detection with standard WGS. The simplified workflow enabled by WGS-based ctDNA detection may thus pave the way to wide-scale clinical application, given the anticipated decline in raw sequencing costs. In addition, the approach presented here requires a lower amount of input material (1 ml of plasma), in contrast to deep sequencing approaches, where sensitivity hinges on completely saturating the available GEs. This aspect is an important consideration in clinical applicability, as sampling-related variability of the amount of available plasma (and GEs) might hinder real-world application of deep targeted approaches, leading to fluctuating sensitivity.

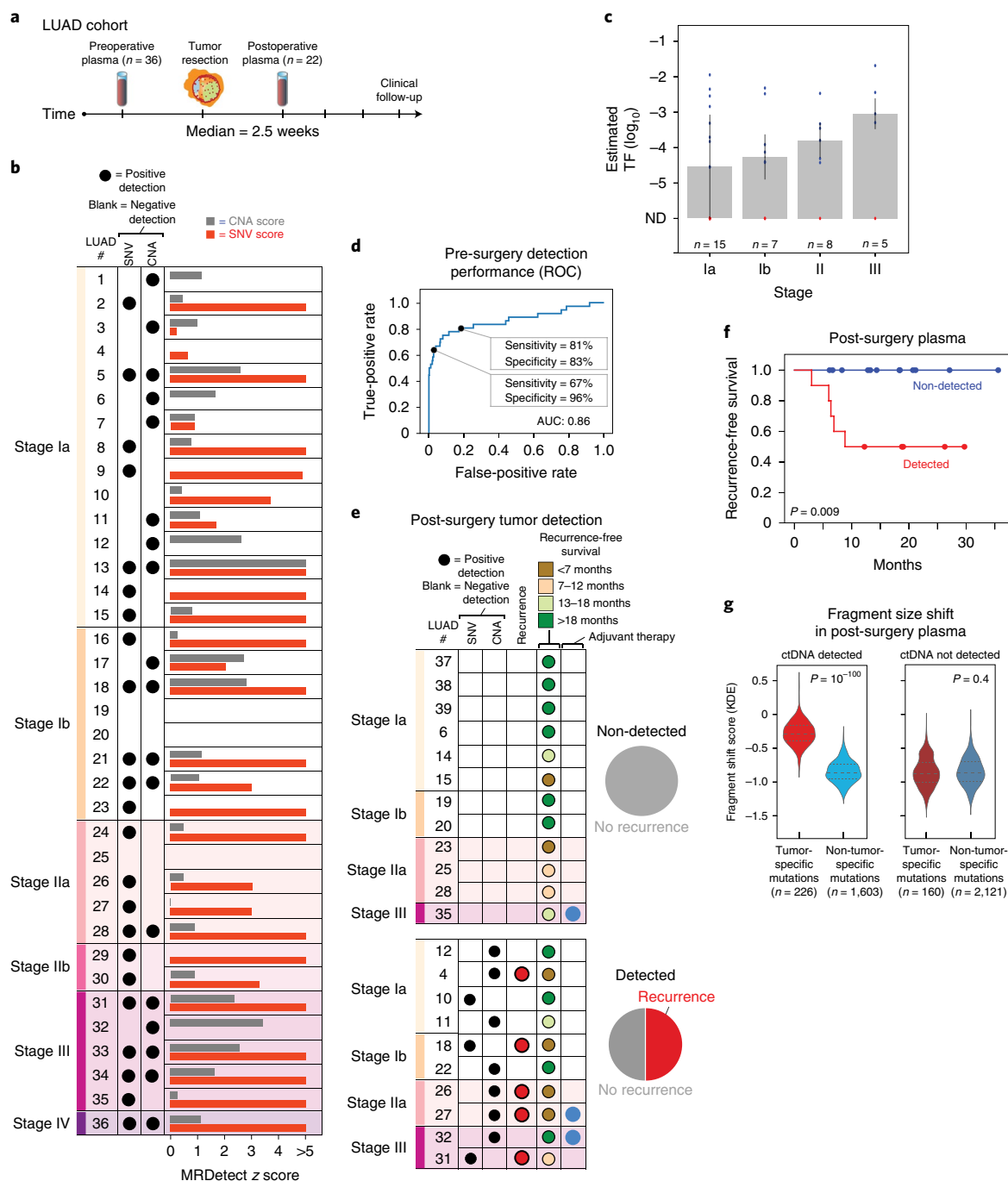
Notably, the application to clinical care will also be closely informed by the negative and positive predictive values. For example, for stage Ib LUAD, the estimated recurrence rate might be as high as 40–50%^{38,39}. In this scenario, the positive predictive value

Fig. 4 | Detection of ctDNA using MRDetect in melanoma during immunotherapy and colon cancer postoperatively. **a**, Error rate estimation in a cohort of test control plasma samples ($n=30$) with and without error suppression. Applying SVM error suppression and paired-end read concordance allow sequencing error reduction by a median of 21-fold and increase the uniformity between samples (two-fold decrease in the coefficient of variation). Box plots represent median, bottom and upper quartiles; whiskers correspond to 1.5× IQR. **b**, Melanoma treatment response during immunotherapy (nivolumab) is monitored by blood samples. Treatment monitoring by computed tomography (CT) shows response to therapy but residual disease after 3 months of therapy (upper panel). MRDetect z scores effectively track tumor responses, matching radiographic changes, in higher temporal resolution than what is feasible with imaging (middle panel). Ichor-CNA sensitivity captures initial treatment response dynamics but does not detect residual disease after 3 months of treatment. Log z score is calculated from a single plasma sample for each timepoint compared to a panel of control samples ($n=30$). **c**, Illustration of the colon cancer clinical cohort and potential clinical utility for liquid biopsy residual disease detection postoperatively. Upon surgical resection of the tumor, the primary tumor is sequenced to define the tumor mutational compendium. Preoperative plasma is used to verify tumor mutational integration detection in cfDNA and establish detection performance metrics. Postoperative plasma (median, 6 weeks) is used to establish the presence of residual disease. **d**, CRC detection on preoperative plasma samples ($n=19$ patients), using SNV (red) and CNA (gray) MRDetect methods. z-score estimation (bar plots) was used to calculate mutation signature detection in the patient plasma in comparison to detection in the control plasma test cohort ($n=30$). z score (>4 and >1.3) was used for the SNV or CNA methods, respectively, to define a positive tumor detection based on ROC optimization on each method. **e**, ROC analysis on a combined detection model of SNV and CNA mutational compendia. Preoperative plasma samples ($n=19$) were used as the positive label, and the panel of control plasma samples against all patient mutational compendia ($n=570$; 19 mutational compendia assessed across 30 control samples) was used as the negative label. Two operational points (z-score thresholds) are shown that correspond with two different specificity regimes. **f**, Detection of ctDNA postoperatively (6 weeks after surgery) in plasma samples ($n=19$) was performed using z-score estimation in comparison to the cohort of control plasma samples ($n=30$). Patients with clinical recurrence are indicated by red dots. Postoperative ctDNA detection (lower panel) is associated with early disease recurrence, as shown in the pie chart. None of patients with the non-detected, postoperative ctDNA (upper panel) showed clinical recurrence after median clinical follow-up of 15 months (range, 4–32). **g**, Kaplan–Meier disease-free survival analysis was done over all patients with detected ($n=7$) and non-detected ($n=12$) postoperative ctDNA. Postoperative ctDNA detection shows association with shorter recurrence-free survival (two-sided log-rank test). MSI, microsatellite instability; MEL, melanoma.

(PPV) and negative predictive value (NPV) of MRDetect (with sensitivity of 81% and specificity of 83%) are 0.8 and 0.84, respectively. With the higher thresholds (sensitivity of 67% and specificity

of 96%), PPV and NPV are 0.93 and 0.78, respectively. These performance metrics are in line with current risk–benefit assessment that translates to practice guidelines offering adjuvant therapy for





stage IIa non-small cell lung cancer with a recurrence rate of 0.55. Although further validation in large cohort studies is required, these proof-of-principle data highlight the potential of MRDetect to deliver precision stratification of patients for adjuvant therapy and, more broadly, to address the urgent unmet need of informing the complex residual disease therapy decision-making.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-020-0915-3>.

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Fig. 5 | Detection of ctDNA using MRDetect in LUAD preoperatively and postoperatively. **a**, Illustration of the LUAD clinical cohort and potential clinical utility for liquid biopsy residual disease detection postoperatively. Upon surgical resection of tumor, the primary tumor is sequenced to define the tumor mutational compendium. Preoperative plasma is used to verify tumor mutational integration detection in cfDNA and establish detection performance metrics. Postoperative plasma (median, 2.5 weeks) is used to establish the presence of residual disease. Patients with detectable ctDNA in their postoperative plasma samples show higher probability for disease recurrence even when there is no other clinical or imaging confirmation at that time. **b**, Early-stage LUAD detection on preoperative plasma samples ($n=36$ patients), using SNV (red) and CNA (gray) MRDetect methods. z score estimation (bar plots) was used to calculate the mutation signature detection in the patient plasma in comparison to detection in the control plasma test cohort ($n=30$). z score was used for the SNV or CNA methods to define a positive tumor detection, indicated by black dots. **c**, Early-stage LUAD tumor fraction estimation per stage. All preoperative plasma samples were considered for TF estimation by MRDetect; non-detected samples were assigned a 0 value. The bar plot represents the median TF value per stage; the black line represents the confidence intervals (IQR); and the specific patient TF values are overlaid for the detected (blue dots) and non-detected (red dots) patients. Estimated TF shows association with cancer stage, with stage III samples showing the highest TF values. **d**, ROC analysis on a combined detection model of SNV and CNA mutational compendia. Preoperative plasma samples ($n=36$) were used as the positive label, and the panel of control plasma samples against all patient mutational compendia ($n=1,080$; 36 mutational compendia assessed across 30 control samples) was used as the negative label. Two operational points (z-score thresholds) are shown that correspond with two different specificity regimes. **e**, Detection of ctDNA postoperative (median, 2.5 weeks after surgery) in plasma samples ($n=22$) was performed using z-score estimation in comparison to the cohort of control plasma test cohort ($n=30$). z score was used for the SNV or CNA methods to define a positive tumor detection, indicated by black dots. Patients with clinical recurrence are indicated by red dots. Postoperative ctDNA detection (lower panel) is associated with early disease recurrence, as shown in the pie chart. None of patients with the non-detected ctDNA (upper panel) showed clinical recurrence after median clinical follow-up of 18 months (range, 6–36). **f**, Kaplan–Meier disease-free survival analysis was done over all patients with detected ($n=10$) and non-detected ($n=12$) postoperative ctDNA. Postoperative ctDNA detection shows association with shorter recurrence-free survival (RFS) (two-sided log-rank test). **g**, Fragments containing tumor mutations show fragment size shift in comparison to non-tumor (cross-patient) mutations in the same patient plasma. The fragment size shift was estimated on detected postoperative patients and non-detected postoperative patients using a kernel density estimation method (Methods). Patients who showed ctDNA detection postoperatively also exhibit significant shift (two-sample t-test) in their tumor-specific mutation detections in the patient plasma (light red, $n=226$), compared to non-tumor (cross-patient) detected mutations on the same samples (light blue, $n=1,603$). Patients who do not show ctDNA detection postoperatively also do not exhibit significant fragment size shift in tumor-specific mutation detections (dark red, $n=160$) compared to non-tumor (cross-patient) detected mutations on the same samples (dark blue, $n=2,121$). Violin plots depict kernel density estimates of the density distribution. Center dashed lines represent the median, and dashed lines represent the IQR. IQR, interquartile range.

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Methods

Human subjects, sample collection and processing. This study was approved by the local ethics committee and by the institutional review board (IRB) and was conducted in accordance with the Declaration of Helsinki protocol. Blood samples were collected in blood collection tubes (Streck) from patient and healthy adult volunteers enrolled in clinical research protocols at NewYork-Presbyterian/Weill Cornell Medical Center, Memorial Sloan Kettering Cancer Center or Massachusetts General Hospital, which were approved by the respective IRBs. Tumor tissues were collected from resected lung and colorectal cancer biopsies. The diagnosis of LUAD and CRC was established according to World Health Organization criteria and confirmed in all cases by an independent pathology review. Informed consent on IRB-approved protocols for genomic sequencing of patients' samples was obtained before the initiation of sequencing studies.

Germline and tumor DNA processing. Tumor tissue and matched germline DNA from peripheral blood mononuclear cells (PBMCs) or adjacent normal tissue were collected and stored at -80°C until they were processed for extraction (Supplementary Table 3). Genomic DNA was extracted from tumor tissue using the QIAamp DNA Mini Kit (Qiagen). Genomic DNA was extracted from PBMCs using the QIAamp DNA Blood Kit (Qiagen). Libraries were prepared using the TruSeq DNA PCR-Free Library Preparation Kit (Illumina) with 1 μg of DNA input after the recommended protocol⁴¹, with minor modifications as described below. Intact genomic DNA was concentration normalized and sheared using the Covaris LE220 sonicator to a target size of 450 bp. After cleanup and end repair, an additional double-sided bead-based size selection was added to produce sequencing libraries with highly consistent insert sizes. This was followed by A tailing, ligation of Illumina DNA Adapter Plate adapters and two post-ligation bead-based library cleanups. These stringent cleanups resulted in a narrow library size distribution and the removal of remaining unligated adapters. Final libraries were run on a Fragment Analyzer (Agilent) to assess their size distribution and quantified by qPCR with adapter-specific primers (Kapa Biosystems). Libraries were pooled together based on expected final coverage and sequenced across multiple flow cell lanes to reduce the effect of lane-to-lane variations in yield. WGS was performed on the HiSeq X (HCS HD 3.5.0.7; RTA v2.7.7) at 2×150 -bp read length, using SBS v3 reagents.

PDX plasma collection. Mouse PDX studies were reviewed and approved by the Institutional Animal Care and Use Committee at Weill Cornell Medicine (protocol no. 2014-0024). PDXs were established using fresh pathologic tissue fragments from a patient with triple-negative breast cancer and implanted subcutaneously into 8-week-old anesthetized NSG mice. Implant growth was assessed by palpation, and, when required, tumor masses were harvested ($<1.5\text{ cm}^3$). Recipient animals were checked regularly and sacrificed at early signs of distress. Serial propagation was performed by implanting tissue seed, as described previously⁴². Blood collection was obtained via tail vein. Blood was collected into Eppendorf tubes containing 5 μl of Streck solution. Samples were spun for 15 min at 14,000 r.p.m. Plasma was collected and stored at -80°C . cfDNA was extracted and sequenced to shallow WGS (1 $\times$) to obtain human (Tumor) circulating DNA and mouse (Normal) circulating DNA.

Plasma DNA processing. At the same day of blood collection, blood collection tubes were centrifuged at 2,000 r.p.m. for 10 min to separate plasma. cfDNA was then extracted from human blood plasma by using the Mag-Bind cfDNA Kit (Omega Bio-Tek). Protocol was optimized and modified to optimize yield. Elution time was increased to 20 min on a thermomixer at 1,600 r.p.m. at room temperature and eluted in 35- μl elution buffer. The concentration of the samples was quantified by a Qubit Fluorometer (Thermo Fisher), and they were run on a fragment analyzer by using the High Sensitivity Genomic DNA Analysis Kit (Agilent) to define the size of cfDNA extracted and genomic DNA contamination.

Extraction protocol optimization. Given the limited amount of cfDNA in the setting of low tumor burden, we examined the potential of cfDNA extraction optimization. To decrease variation derived from sample acquisition and inter-individual variation, we compared available extraction methods on uniform cfDNA material generated through large-volume plasma collections ($\sim 300\text{ cm}^3$) through plasmapheresis of healthy individuals and patients with blood cancer in remission who were undergoing hematopoietic stem cell collection. The large volume of plasma allows the testing of multiple methods and protocol parameters on the same cfDNA input, enabling accurate measurement of subtle differences in yield and quality. We compared cfDNA extraction methods from Capital Biosciences, Zymo Research, Omega Bio-Tek and NeoGeneStar. Extractions were performed on 1 ml of the large-volume plasma sample according to the manufacturer's instructions, and multiple plasma aliquots were processed in parallel to assess both inter- and intra-method variability. The yield and purity of each recovered cfDNA sample was determined using a Qubit Fluorometer (total mass), NanoDrop (Thermo Fisher) via UV absorbance (detection of salt and protein contaminants) and on-chip electrophoresis (size distribution and genomic DNA (gDNA) contamination). We found that the Mag-Bind cfDNA Kit from Omega Bio-Tek displayed overall somewhat better performance (Extended Data

Fig. 1c,d) and ease of use compared to other tested methods. We therefore elected to perform Omega Bio-Tek extraction for patient and control plasma samples.

Plasma cfDNA library preparation and sequencing. Two library preparation methods were used to sequence cfDNA: KAPA Hyper Library Preparation and NEXTflex Cell Free DNA-Seq Kit. We transitioned to the KAPA method owing to the ability to generate a library on variable input material and somewhat lower duplication rate. We note no significant differences between the methods in terms of MRDetect performance and coverage variation across the genome. For KAPA Hyper Library Preparation, samples having a mass above 1 ng were prepared for next-generation sequencing on Illumina's HiSeq X by using a modified manufacturer's protocol. The protocol was scaled down to half reaction by using 25 μl of extracted cfDNA. IDT for Illumina TruSeq Unique Dual Indexes were used by diluting 1:15 with EB, and ligation reaction was adjusted to 30 min. Additional 0.8 $\times$ SPRIselect magnetic beads (Beckman Coulter) cleanup was included to the protocol after post-ligation cleanup to remove excess adapters and adapter dimers. cfDNA from 1 ml of plasma was used for all of the plasma samples in this study. For samples with low concentration (five samples in the entire study), an additional 1 ml of plasma was extracted, and the DNA aliquot with the highest mass was used for library preparation (see Supplementary Table 3 for the total plasma extracted for each sample). The number of PCR cycles was dependent on initial cfDNA total mass. For samples with more than 5 ng of total cfDNA, five PCR cycles were performed. For samples with less than 5 ng of total cfDNA, 7–10 PCR cycles were performed. Quality metrics were performed on the libraries by Qubit Fluorometer, High Sensitivity DNA Analysis Kit and KAPA SYBR FAST qPCR Kit (Kapa Biosystems).

For the NEXTflex Cell Free DNA-Seq Kit, 1–5 ng of the extracted cfDNA were prepared to make libraries for next-generation sequencing on Illumina's HiSeq X by using the manufacturer's protocol for Illumina sequencing. NEXTflex DNA Barcode was diluted 1:8, and ten PCR cycles were performed on the libraries. Additional 0.8 $\times$ SPRIselect magnetic beads cleanup was included to the protocol after post-PCR cleanup to remove excess adapter dimers. Quality check was performed by Qubit Fluorometer, High Sensitivity NGS DNA Analysis Kit by AATI and KAPA SYBR FAST qPCR Kit. WGS was performed on the HiSeq X (HCS HD 3.5.0.7; RTA v2.7.7) at 2×150 -bp read length, using SBS v3 reagents.

Preprocessing, quality control analysis and sample identification and concordance.

WGS reads for primary tumor, matched germline and plasma samples were demultiplexed using Illumina's bcl2fastq (v2.17.1.14) to generate FASTQ files. The primary tumor and matched germline WGS were submitted to the New York Genome Center somatic preprocessing pipeline, which includes alignment to the GRCh37 reference (1000 Genomes version) with BWA-ALN (v0.6.2)⁴³. We used a modified alignment pipeline for plasma cfDNA to accommodate adapter trimming. We did this after observing increased adapter contaminated reads in cfDNA samples as compared to tumor samples, presumably due to the fact that cfDNA has shorter fragment size, which can lead to R1 and R2 overhang (data not shown). For cfDNA samples, we used Skewer⁴⁴ for adapter trimming (default settings) and subsequently aligned samples using BWA-MEM (default settings) to the GRCh37 reference (1000 Genomes version). For all samples, duplicate marking and sorting was done using NovoSort MarkDuplicates (v1.03.01, a multi-threaded bam sort/merge tool by Novocraft Technologies; <http://www.novocraft.com>), followed by indel realignment (done jointly for the tumor and matched germline) and base quality score recalibration using GATK (v3.4.0; <https://software.broadinstitute.org/gatk>), resulting in a final coordinate sorted bam file per sample. Alignment quality metrics were computed using Picard (v1.83; QualityScoreDistribution, MeanQualityByCycle, CollectBaseDistributionByCycle, CollectAlignmentSummaryMetrics, CollectInsertSizeMetrics, CollectGcBiasMetrics and CollectOxoGMetrics) and GATK (average coverage, percentage of mapped and duplicate reads). To specifically assess for potential sample contamination, we applied Conpair⁴⁵, which validated genetic concordance among the matched germline, tumor and plasma samples, as well as evaluated any inter-individual contamination in the samples. Samples that showed low concordance (<0.99) were excluded from further analysis. Specifically, three preoperative plasma samples from LUAD patients 37, 38 and 39 were rejected from analysis owing to low concordance score.

Generation of synthetic-plasma DNA admixtures. We generated in silico admixtures of tumor and matched germline DNA from eight patients with various cancer types, including LUAD, breast cancer, melanoma and osteosarcoma (Supplementary Table 1). Admixtures were generated by downsampling and mixing, using SAMtools (v1.1, view -s and merge commands), of tumor and germline reads sequenced with the same protocol and sequencing device (see tumor DNA processing above) used for the LUAD cohort reported in our study. The ratio of tumor versus germline reads was defined by the TF, taking into account the tumor-specific purity and ploidy using the following model. In the tumor sample, the fraction of DNA that is actually coming from the tumor (T_{TF}) can be characterized by the following equation:

$$T_{\text{TF}} = \frac{T_{\text{reads}}}{T_{\text{reads}} + N_{\text{reads}}} = \frac{P_U \cdot P_L}{P_U \cdot P_L + (1 - P_U) \cdot 2} \quad (1)$$

where P_U denotes purity, P_L denotes ploidy in the tumor sample, T_{reads} denotes the number of tumor tissue reads sequenced from the tumor sample and N_{reads} denotes the number of healthy tissue reads (normal tissue in the tumor sample or contamination) sequenced from the tumor sample. Downsampling the original tumor to a lower TF (so as to represent highly tumor-diluted plasma samples) will require the downsampling ratio S :

$$S = \frac{TF_{required}}{T_{TF}} = TF_{required} * \frac{P_U * P_L + (1 - P_U) * 2}{P_U * P_L} \quad (2)$$

Lastly, scaling down the coverage of each sample, tumor and germline and mixing will take the following form:

$$tumor\ read\ ratio = S * \frac{cov_{req}}{cov_T} \quad (3)$$

$$germline\ read\ ratio = (1 - S) * \frac{cov_{req}}{cov_N}$$

where cov_{req} is the required read depth coverage for the admixture sample and cov_T , cov_N are the read depth coverage of the tumor and germline samples, respectively.

Tumor / Normal somatic mutation calling. The primary tumor and matched germline bam files were processed through somatic variant calling pipeline that consists of MuTect (v1.1.7)²³, Strelka (v1.0.14)⁴⁶ and LoFreq (2.1.3a)⁴⁷ for SNVs. We note that, for the two patients with metastatic melanoma, no primary tumor was available for sequencing. We therefore performed mutation calling comparing the pre-treatment plasma WGS to PBMC WGS. To achieve stringent somatic variant calling, we enforced intersection between callers. We further removed triallelic sites and common germline variants (minor allele fraction (MAF) > 5% in cancer genes *DNMT3A*, *TET2*, *JAK2*, *ASXL1*, *TP53*, *GNAS*, *PPM1D*, *BCORL1* and *SF3B1* and with MAF ≥ 1% elsewhere in the genome, as reported in the 1000 Genomes Project release 3 and the Exome Aggregation Consortium server). Finally, we removed a subset of artifactual calls by the use of a 'blacklist' generated by calling somatic variants on 16 random pairings of 80×/40× in-house-sequenced HapMap WGS data. Small deletions and insertions (indels) were not used in our methodology owing to smaller frequency and somewhat greater challenges in calling accuracy. CNAs, including deletions, amplifications and copy-neutral loss of heterozygosity, were called using NBIC-seq (v0.7)⁴⁸. Segments with $\log_2$ greater than 0.2 were categorized as amplifications, and segments with $\log_2$ less than -0.235 were categorized as deletions (corresponding to a single copy gain or loss, respectively, at 30% purity genome).

Plasma cfDNA SNV identification. Sensitive detection of patient-specific compendia of SNVs was performed by searching the plasma WGS for all sites from the matched patient-tumor compendium with corresponding mutations in the same genomic site and the same substitution. To efficiently identify variants present in the sequencing data, we used a custom Python script (Python version 3.6.2), which uses the pysam module to efficiently extract alignments harboring variants and extracted any read that both uniquely maps to a variant of interest and was in an aligned portion of the read (no clipping or soft masking at the position of the variant). All extracted reads were then subjected to the SVM (see below) model for further classification.

Sequencing error suppression. The detection of mutations in cfDNA WGS required a new conceptualization of the mutation-calling process. Current mutation calling is typically 'locus centric'^{23,46,47}, whereby the presence of multiple supporting reads for the same mutation is incorporated into statistical models. However, in highly underrepresented ctDNA detection in low-burden disease, the tumor fraction is orders of magnitude lower than the inverse of depth of sequencing, such that, at best, only a single supporting read is available. We therefore generated a new analytic framework that is 'read centric'⁴⁹, using a machine learning model to scan through individual reads and provide a measure of confidence to classify them as likely true somatically mutated ctDNA versus sequencing artifacts. These data were then integrated across the genome using analytic models (see below) to measure the tumor fraction representation in cfDNA.

Our read-centric de-noising method applies an SVM framework. Five features, which are known to represent sequencing error patterns, were included in the model training and which showed association to artifactual detections in our training control plasma cohort ($n = 8$). Variant base quality (VBQ) indicates our trust in the particular mismatch, which showed significant enrichment with sequencing error ($P < 10^{-100}$, two-sample t -test; Extended Data Fig. 2b,c). Mean read base quality (MRBQ) represents the overall quality of sequencing within a particular read pair. Position in read (PIR) captures cycle-specific errors, as 3' showed higher association with sequencing error (Extended Data Fig. 2d). R1 and R2 allow us to test concordance between overlapping read-pair sequences, where discordance is associated with sequencing error (Extended Data Fig. 5a). Mapping quality is an alignment-provided metric corresponding to the trust the aligner has in the particular alignment.

To train the read-centric SVM model, we first focused on building a high-quality truth set of confident alterations and likely errors. For confident alterations, we aimed for variants with a high degree of support. We used GATK (v3.4.0) to call variants on each of the training control plasma samples, using the -L flag to specify only returning sites in dbSNP (build 151) variants. For error mutations, we searched for mutations with a low degree of support by performing pileup using SAMtools mpileup to identify mismatches across the genome. These were then filtered for coverage (coverage > 10×) while enforcing that the variant has low support of ≤ 0.1 VAF. We sampled 10,000 reads randomly from the training set of eight control patients' cfDNA for each class for training.

For model training, we used sklearn's SVM package, using linear SVM with $C: 1.0$, tol 1e-7, hinge loss and l2 regularization. Using 0 as a cutoff, we evaluated model performance by F1, sensitivity and specificity using ten-fold cross validation (Extended Data Fig. 2e). Throughout development, we considered the decision boundaries when various features were fixed. Visualizations were interpreted for expected relationships between features: VBQ correlates with PIR; R2 has a more stringent bias; MRBQ correlates with VBQ; and, when both VBQ and MRBQ are high, we make a confident decision. As a comparison to our SVM method, we evaluated the performance of a random forest fit using the same data set and features. The number of trees was set to 50. The SVM model outperformed the random forest model across several metrics (Extended Data Fig. 2e).

Additional de-noising methods. Paired-end sequencing of the short cfDNA fragments (read length, 150 bp) results in overlapping R1 and R2 reads, which provide an additional means of error suppression due to overlapping information from the same DNA fragment. Given the low probability of a sequencing error occurring in both of the overlapping reads, pairs that contained the same alternative base in their overlaps (concordant mutation) were considered as more likely to represent true mutations as opposed to pairs in which only one read differed (discordant), which are more likely to reflect sequencing errors. To test this notion, we evaluated the proportion of concordant read pairs at dbSNP sites and at likely artifactual sites, which were defined by read pairs with variants overlapping the union of all patient-somatic SNV compendia across all patients with LUAD who were observed with the same variant in the control plasma samples (Extended Data Fig. 5a). This analysis revealed that pairs are far more likely to be discordant at artifactual sites and mostly concordant at sites containing true germline variant. This analysis was conducted using a custom Python script (Python version 3.6.2), which uses the pysam module. Briefly, for each variant, reads containing the variant of interest and their read mates were compared to evaluate whether both (concordant) or only one (discordant) contained the variant of interest, and discordant read pairs were discarded. Additionally, we generated a 'blacklist' of regions in the genome that are more likely to contain artifactual variants. To do this, we performed artifact detection mode on the training control plasma samples ($n = 8$); any variant detected in at least two controls was included in the blacklist and removed from subsequent patient plasma analysis.

Plasma SNV-based ctDNA detection and quantification. We reasoned that the fraction of the patient-specific SNVs that are observed in the plasma cfDNA WGS follows a binomial distribution over N independent Bernoulli trials, where N is the number of SNVs in the patient-specific mutational compendium (identified through standard SNV detection on matched tumor and germline DNA WGS data). Each such trial includes multiple rounds of random sampling that depends on the local coverage, where the probability of sampling a DNA fragment containing a given variant in each round is defined as the TF (the fraction of circulating tumor DNA in the cfDNA pool). We note that we have not explicitly modeled heterozygosity or lower VAF due to subclonal events, and thus our Bernoulli model likely somewhat underestimates the true TF. However, these aspects will likely lead to a two-to-three-fold underestimation at most (given depth of sequencing, purity for subclonal SNVs), which is anticipated to have limited effect considering the heavy dilution of ctDNA in low-burden disease (<1%).

Therefore, the relationship among coverage, mutation load (SNV per tumor) and the number of detected variants in cfDNA WGS and the TF corresponds to the following equation:

$$M = N(1 - (1 - TF)^{cov}) + \mu * R \quad (4)$$

where M denotes the number of SNVs detected in the plasma sample, N denotes the number of SNVs (mutation load) in the patient-specific mutational compendium, TF denotes the tumor fraction, cov denotes the local coverage in sites with a tumor-specific SNV, μ denotes the mean noise rate (number of errors/number of reads evaluated) that corresponds to the patient-specific SNV compendium evaluated in control plasma WGS data (see below) and R denotes the total number of reads covering the patient-specific mutational compendium. This relationship allows the calculation of the plasma TF from the mutation detection rate, even in extremely low allele fraction where the mutation allele fraction itself is not informative (random sampling between 0 and 1 supporting read at best).

To address variation in sequencing artifact noise (μ) across patients with different mutational compendia, we applied the patient-specific mutational

compendium to calculate the expected noise distribution across the cohort of control plasma samples. The process described above is performed to detect the patient-specific SNVs in control plasma samples or other patients (cross-patient analysis). These detections represent the background noise model for which we calculate the mean and s.d. (μ, σ) of artifactual mutation detection rate. Confident cfDNA tumor detection can then be defined by converting the patient-specific detection rate ($\text{det_rate} = \text{number of SNVs detected in cfDNA} / \text{number of reads checked} = M/R$) to a z score $= \frac{\text{det_rate} - \mu}{\sigma}$, and defining a threshold that will keep the specificity above 80% (z score > 1.2). Specificity and sensitivity performance values were further validated using ROC curve using the Python (version 2.7.10) module `numpy.metrics.roc_curve` and `numpy.metrics.roc_auc_score`.

Calculating the patient TF from point mutation detection was then carried out by the following equation (which is an inversion of Eq. 4):

$$TF = 1 - (1 - [M - \mu^*R]/N)^{1/cov} \quad (5)$$

where M denotes the number of SNVs detected in the plasma sample, N denotes the number of SNVs (mutation load) in the patient-specific mutational compendium, TF denotes the tumor fraction, cov denotes the local coverage in sites with a tumor-specific SNV, μ denotes the noise rate (number of errors/number of reads evaluated) that corresponds to the patient-specific SNV compendium and R denotes the total number of reads covering the patient-specific mutational compendium.

Plasma CNA-based quantification. To generate a patient-specific CNA compendium (amplifications and deletions), we applied NBIC-seq⁴⁸ to the primary tumor and generated an interval list of amplifications, deletions and copy-neutral loss of heterozygosity. This interval list was then interrogated in cfDNA WGS to calculate an integrated signal (score) for representation of ctDNA in the cfDNA. Average coverage per base pair was calculated in plasma samples using GATK (v3.4.0) `DepthOfCoverage`. The genomic region of interest (CNA or Neutral segments) was divided into non-overlapping 500-bp bins, and median depth of coverage per bin was calculated. We note general stability of performance for bin size between 200 bp and 1 Kbp, with lower performance with increasing bin size (data not shown). To compare across samples with varying depths of coverage, we normalized the depth of coverage in each bin by dividing the coverage by the average coverage of the sample. To further correct for sample-specific variation in depth of coverage, a robust z -score normalization was applied to each plasma sample separately. Median and median absolute deviation (MAD) were calculated per sample over the entire genomic region of interest (over the aggregate of all the patient's CNA segments), and subsequently all genomic bins were normalized by subtracting the median and dividing the result by the MAD. Specifically, every 500-bp bin was converted to a $\text{normalized_coverage} = \frac{b_i - \text{Median}}{\text{MAD}}$.

Generating plasma reference sample. Read depth profiles from cfDNA and from gDNA (tumor or PBMC) showed significant differences (Extended Data Fig. 5b,c), possibly due to differences in library preparation (PCR versus PCR-free protocols) and non-uniformity of cfDNA coverage driven by DNA degradation and epigenetic features (for example, chromatin accessibility). To ensure that the signal calculated is not driven by these effects, we generated a control plasma reference for plasma differential coverage measurement. Control plasma training set samples ($n = 8$) were randomly downsampled and admixed, using SAMtools (v1.1, `view -s, merge`), to generate a reference cfDNA plasma sample with 25 $\times$ depth of coverage. The robust z -score-normalized coverage of this reference sample was used as the reference plasma for all subsequent analysis, in place of patient-specific PBMC reference. We further note that detection results with the plasma reference and with patient-specific PBMC WGS showed high concordance, confirming the robustness of these results (Supplementary Table 4).

Removing artifactual CNA events. We observed that some genomic regions show recurrent amplification and deletion patterns in control samples; these include centromeres and other regions of poor coverage known as 'gap regions'. To remove genomic bins (500 bp) with recurrent and significant high or low coverage compared to the rest of the genome, we removed bins with extreme coverage values ($> 1.5 \text{ MAD}$), on the reference sample, after robust z -score normalization. Given that each sample was diluted in the plasma reference (composed of downsampled eight samples), this likely resulted in removal of only recurrent artifactual amplification or deletion. Somatic CNA events originating from possible clonal hematopoiesis can also create biases in plasma cfDNA CNA analysis, as most cfDNA is derived from blood cells. To assess for this potential artifactual, CNA-matched PBMC WGS was assessed for CNAs using NBIC-seq (v0.7) against a contemporary normal sample, NA12878 (<http://www.internationalgenome.org/data-portal/sample/NA12878>), sequenced using the same protocol as the PBMC. Copy-neutral loss of heterozygosity events were also examined using B-allele frequency analysis. CNAs were called using NBIC-seq (v0.7)¹¹ on PBMC versus contemporary normal. Segments (> 1 Mb in length) with $\log_2$ greater than 0.2 were categorized as amplifications, and segments with $\log_2$ less than -0.235 were categorized as deletions (corresponding to a single copy gain or loss, respectively, at 30% purity genome) in the PBMC. These regions were then excluded from the

patient-specific CNA interval list. Detected events were removed from the CNA and neutral interval lists to prevent biases in MRDetect CNA signal. Three patients had detectable somatic PBMC events: LUAD10 (amp Chr12:60138-133841502), LUAD26 (CN-LOH Chr4:50400000-191044164) and CRC03 (del Chr3:234305-80851349; del Chr5:75605307-180877637; del Chr7:95649215-159128563; del Chr10:50003039-108417985; del Chr15:36365636-63901029; del Chr15:36365636-63901029; del Chr17:7602691-20374289; del Chr18:24227106-78017148).

Plasma cfDNA CNA signal measurement. Differential coverage between plasma sample (pre-surgery or post-surgery) and the reference showed a tendency for a positive skew in bins overlapping patient-specific amplifications and a negative skew in bins overlapping patient-specific deletions. Therefore, multiplying the differential (Plasma_sample—PON_reference) depth of coverage by the directionality of the CNA segment (amplification multiplied by +1, deletion multiplied by -1) and summing these values across the CNA regions provides the CNA signal, whereas neutral regions (diploid regions in the matched tumor) provide the background noise, in which random directionality in depth of coverage is expected to result in low integrated signal. This is represented by the following equation:

$$\text{CNA Signal} = \sum_{i=1}^M [(P(i) - N(i)) * \text{sign}(T(i) - N(i))] \quad (6)$$

where M is the number of non-overlapping 500-bp bins covering the region of interest (CNA segment). $P(i)$ and $N(i)$ are robust z -score-normalized depth of coverage values in the window i for the plasma sample and reference, respectively. $\text{Sign}(T(i) - N(i))$ represents the direction of the tumor CNA segment (amplification multiplied by +1, deletion multiplied by -1) that overlap with window i .

To address variation in noise across patients with different CNA compendia, we used the patient-specific CNA interval list to calculate patient-specific noise distribution over a cohort of control plasma samples ($n = 30$). The process described above was done to detect the patient-specific CNA compendia in control plasma samples or in other patients' plasma (cross patient). These detections represent the background noise model for which we calculated the mean and s.d. (μ, σ) of artifactual background noise. Subsequently, the patient-integrated CNA signal and control noise values were converted to z scores to define the significance of CNA detection in patient samples compared to background noise.

These detections represent the background noise model for which we calculated the mean and s.d. (μ, σ) of artifactual CNA detection signal. Confident cfDNA tumor detection can then be defined by converting the patient-specific CNA signal to a z score $= \frac{\text{CNA_signal} - \mu}{\sigma}$, and defining a threshold that will keep the specificity above 80% (z score > 1.2). Specificity and sensitivity performance values were further validated using ROC curve using the Python (version 2.7.10) module `numpy.metrics.roc_curve` and `numpy.metrics.roc_auc_score`.

Using the CNA mixture model on copy-neutral regions of the same tumor genome was done as an additional negative control with the same model defined above (Eq. 6). In this case, for each 500-bp bin, the directionality was defined as the local sign ($T - N$), meaning the local sign of the difference between the tumor-normalized coverage and the germline-normalized coverage in this specific 500-bp bin (in copy-neutral regions this sign takes random $+1/-1$ values without a defined trajectory).

CNA downsampling. To demonstrate sensitivity at various TFs and size of genome altered, we used the *in silico* admixtures of tumor and matched germline DNA from breast cancer (described in Methods above) and randomly downsampled the CNAs' footprint. We generated eight replicates across TFs (1×10^{-3} to 5×10^{-1} and control) and three replicates across CNA footprint sizes (5–1,050 Mb). We applied MRDetect CNA (described in Methods above) comparing the replicates to control plasma samples to generate CNA z scores and finally compute the median z score across all replicates.

CNA load across tumor types analysis. We applied a junction balance analysis algorithm (JaBba)⁵⁰ on 588 The Cancer Genome Atlas WGS samples across ten tumor types to obtain purity and ploidy corrected CNA segments. Briefly, JaBba⁵⁰ infers somatic segmental copy number simultaneously with rearrangement junction copy numbers from paired tumor-normal WGS. High-density (200-bp bins) coverage data are collected from tumor and normal WGS aligned reads, corrected for GC content and mappability, and rearrangement junctions are inferred by SvABA (<https://doi.org/10.1101/gr.221028.117>). JaBba⁵⁰ reconstructs a junction-balanced genome graph by optimizing a complexity-regularized likelihood function with mixed-integer programming.

Integrating SNV and CNA scores. For definition of tumor detection (preoperatively and postoperatively), we applied thresholds designed based on the ROC analysis. The threshold for SNV detection was defined as the first value to provide $> 95\%$ specificity, z score > 3 for lung (Fig. 5b,e and Extended Data Fig. 9) and z score > 4 for CRC (Fig. 4d,f and Extended Data Fig. 8). The threshold for CNA detection was defined as the first value to provide $> 80\%$ specificity, z score > 0.9 for lung (Fig. 5b,e and Extended Data Fig. 9) and z score > 1.3 for

CRC (Fig. 4d,f and Extended Data Fig. 8). To further evaluate performance, we also applied identical z-score thresholds across the cohorts. For example, in SNV analysis, a z score of >3, selected based on specificity >90% in CRC (specificity of 98%, sensitivity of 90%), yields similar results in LUAD (specificity of 95%, sensitivity of 67%). For the CNA analysis, a z score of >1.3, selected based on specificity >90% in CRC (specificity of 92%, sensitivity of 40%), yields similar results in LUAD (specificity of 89%, sensitivity of 27%). These thresholds did not affect the association between postoperative detection and adverse outcome ($P = 0.03$ for the CRC cohort, $P = 0.024$ for the LUAD cohort).

Additionally, as the SNV noise model and CNA noise model are largely independent, generated by different noise mechanisms, the SNV and CNA z-score values of control plasma samples can be combined together and still generate a normal distribution around zero mean. Measuring this combined noise z score across all control samples by applying all the different patient mutation compendia (total, $n = 132$ combinations) yielded a mean z score of 0.07. In contrast, patient cfDNA CNA and SNV z scores are correlated due to the underlying circulating tumor DNA detection signal, which will allow an increased signal to noise for combined CNA and SNV detection signal. SNV z score and CNA z score are combined to get the integrated score ($\text{integrated_score} = \text{SNV_zscore} + \text{CNA_zscore}$) for all patient plasma and control plasma; the combined metrics performance is shown for lung and CRC (Figs. 4e and 5d).

ichorCNA. ichorCNA¹⁰ (version 2.0) was used as an orthogonal CNA-based method for the estimation of plasma TF in the in silico admixtures and in patient samples. We optimized the input setting for more sensitive detection in low-tumor-burden disease using the modified flags -altFracThreshold 0.001, -normal .99, -NORMWIG for using matched normal instead of the ichor panel of normal; all other settings were set to default values. As a matched normal for the in silico admixtures, we used an independent matched germline sample. For the patient samples, we used our control plasma PON as the matched normal.

Fragment size KDE. Circulating DNA fragments that originate from the tumor show shorter fragment size in comparison to 'normal' DNA fragments that originate mainly from apoptosis of hematopoietic cells (immune cells)^{32,51}. We therefore reasoned that shorter fragment length of DNA fragments of somatic variants detected in cfDNA could provide orthogonal support to the fact that these indeed represent ctDNA. To generate a robust model that can quantify the probability of a single DNA fragment to be from tumor or normal origin, we used a joint KDE to characterize the fragment size distribution of circulating DNA. This method allows to rigorously quantitate the fragment size shift across the entire fragment sizes distribution (80–600 bp), instead of focusing only on <150-bp fragments³⁰. To train the KDE estimator, we used a triple-negative breast cancer PDX to create a purified set of human circulating tumor DNA. We collected blood from the mouse and sequenced the cfDNA. Circulating tumor DNA fragments were then isolated by retaining only human-aligned DNA reads, which were then used as a tumor-labeled training set ($n \sim 3$ M fragments). Normal cfDNA training set was defined by a random sample of reads from our human control plasma cohort ($n \sim 6$ M fragments, from eight control plasma samples). A KDE model was trained over the tumor and normal cfDNA sets using the Python (version 2.7.10) module `scipy.stats.gaussian_kde`, using `scott-rule` to choose optimal bandwidth (`bw_method='scott'`). We then scored collections of cfDNA fragments (for example, tumor-specific detections or cross-patient detection) based on their fragment sizes using the following equation:

$$\text{score} = \text{median}[\log(\text{pdf_tumor}(\text{collection}))] - \text{median}[\log(\text{pdf_normal}(\text{collection}))] \quad (7)$$

where for each specific DNA fragment the $\log\text{pdf}$ value was calculated using `scipy.stats.gaussian_kde().logpdf()` for both the tumor KDE and the normal KDE separately.

The KDE score was tested on the aggregated tumor-specific mutation detections and aggregated cross-patient mutation detection in different patient sets (pre-surgery plasma, ctDNA-positive post-surgery plasma and ctDNA-negative post-surgery plasma). We used a permutation test to check the significance of the difference in KDE score between tumor-specific mutation detection and cross-patient mutation detection. For each mutation set, we randomly subsampled 150 DNA fragments 1,000 times with replacement and calculated the KDE score for each permutation subsample. Violin plot statistics of these permuted KDE score cohorts is shown (Fig. 5g and Extended Data Fig. 10j), and statistical difference was calculated using two-sample t -test.

Reanalysis of targeted panel cfDNA sequencing data. Supplementary Tables 3 and 7 from Phallen et al.¹⁸ were downloaded, and, for each patient in Supplementary Table 3, the VAF information was retrieved from Supplementary Table 7. Distributions were fit to the patients' maximum VAF values using the R package 'fitdistrplus', setting the 'distr' parameter to 'beta' or 'lnorm' as appropriate (Fig. 1b). Goodness of fit was determined by computing the mean squared error between the predicted distributions and the empirical cumulative distributions. The number of GEs (unique coverage) for each patient was taken from

Supplementary Table 4. Linear regression was performed using the 'lm' function in R (version 3.4.2), and R^2 and P value statistics were calculated (Fig. 1c).

Statistical analysis. Statistical analysis was performed with Python 2.7.13 and R version 3.4.2. Continuous variables were compared using Student's t -test, the Wilcoxon rank-sum test or the nonparametric permutation test, as appropriate. All P values are two sided and considered significant at the 0.05 level, unless otherwise noted.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Sequence data has been deposited at the European Genome-phenome Archive (EGA), which is hosted by the EBI and the CRG, under accession number [EGAS00001004406](https://ega-archive.org/studies/EGAS00001004406).

Code availability

The analytic code used for this work is provided for non-commercial use at: <https://ctl.cornell.edu/technology/mrdetect-licence-request>.

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Author contributions

D.A.L., A.Z. and N.K.A. conceived and designed the project. K.P., A.Z., S.R., D.R., G.G., J.R., N.D.O., C.A., M.M., C.F.S., S.K., S.F., G.G.I., V.A., B.H.-L., J.L., N.K.A., A.W., P.W., A.M.R., M.K.C., D.F., T.S., B.M., T.K., J.D.W., G.B. and D.A.L. performed patient selection and prepared samples for sequencing. A.Z., R.C.S., M.S., S.T.K.H., A.J.W., S.D., P.O.B., A.H.L., D.M., G.H., K.Y.H., W.L., C.S., C.C.K., F.G., N.R. and D.A.L. performed the computational genomics analyses. A.Z., R.C.S. and D.A.L. wrote the manuscript with comments and contributions from all authors.

Competing interests

D.A.L., A.Z., V.A. and S.T.K.H. submitted two patent applications. D.A.L. and A.Z. are co-founders of C2i Genomics. D.A.L. participated in an advisory board for Illumina

and has received research support. J.D.W. consulted for Adaptive Biotech, Advaxis, Amgen, Apricity, Array BioPharma, Ascentage Pharma, Astellas, Bayer, Beigene, Bristol Myers Squibb, Celgene, Chugai, Elucida, Eli Lilly, F Star, Genentech, Imvax, Janssen, Kleo Pharma, Kyowa Hakko Kirin, Linneaus, MedImmune, Merck, Neon Therapeutics, Northern Biologics, Ono, Polaris Pharma, Polynoma, Psioxus, Puretech, Recepta, Takara Bio, Trieza, Turvax, Sella Life Sciences, Seramatrix, Surface Oncology, Syndax and Syntalogic. J.D.W. also received research support from Bristol Myers Squibb, Medimmune, Merck Pharmaceuticals and Genentech. J.D.W. holds equity in Potenza Therapeutics, Tizona Pharmaceuticals, Adaptive Biotechnologies, Elucida, Imvax, Beigene, Trieza and Linneaus.

Additional information

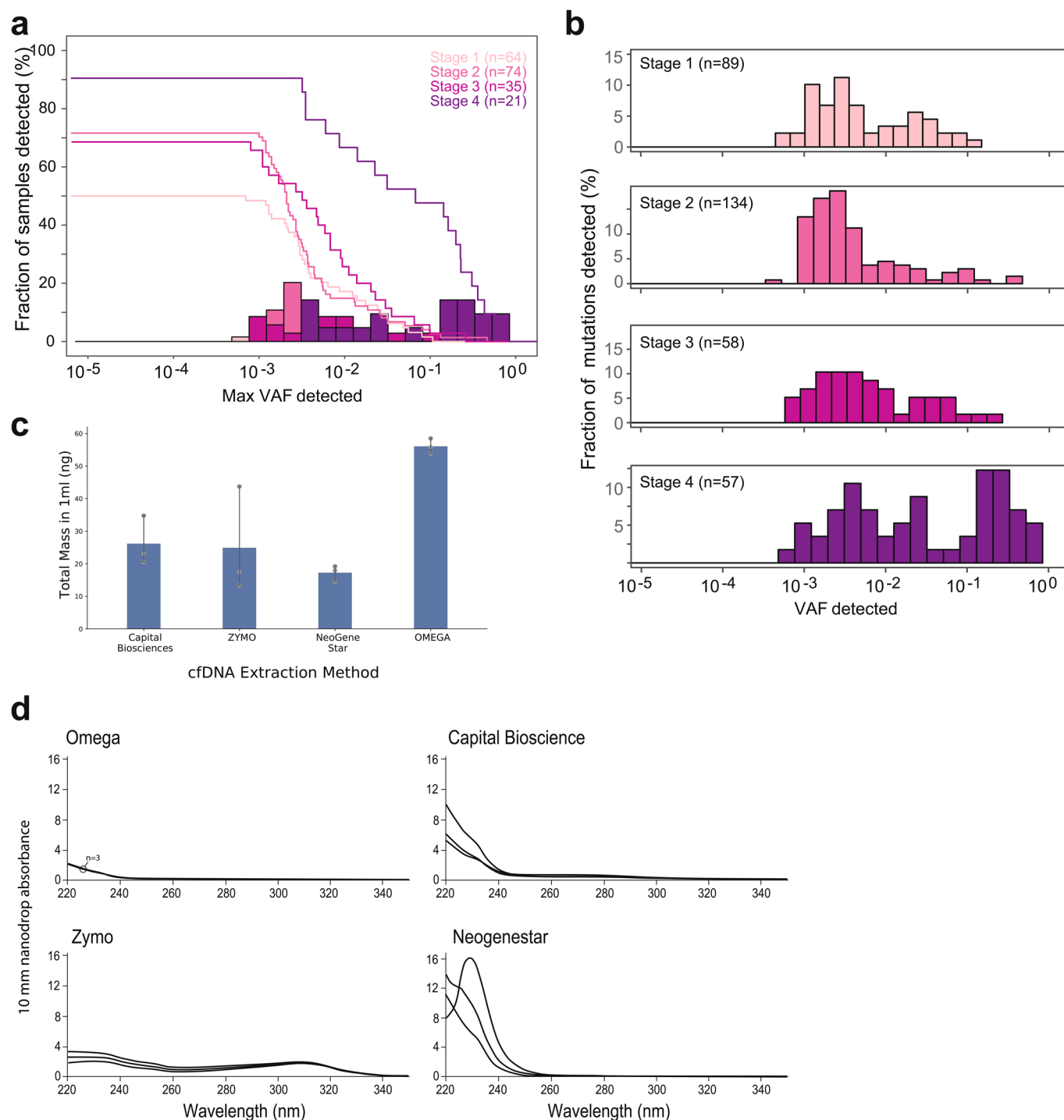
Extended data is available for this paper at <https://doi.org/10.1038/s41591-020-0915-3>.

Supplementary information is available for this paper at <https://doi.org/10.1038/s41591-020-0915-3>.

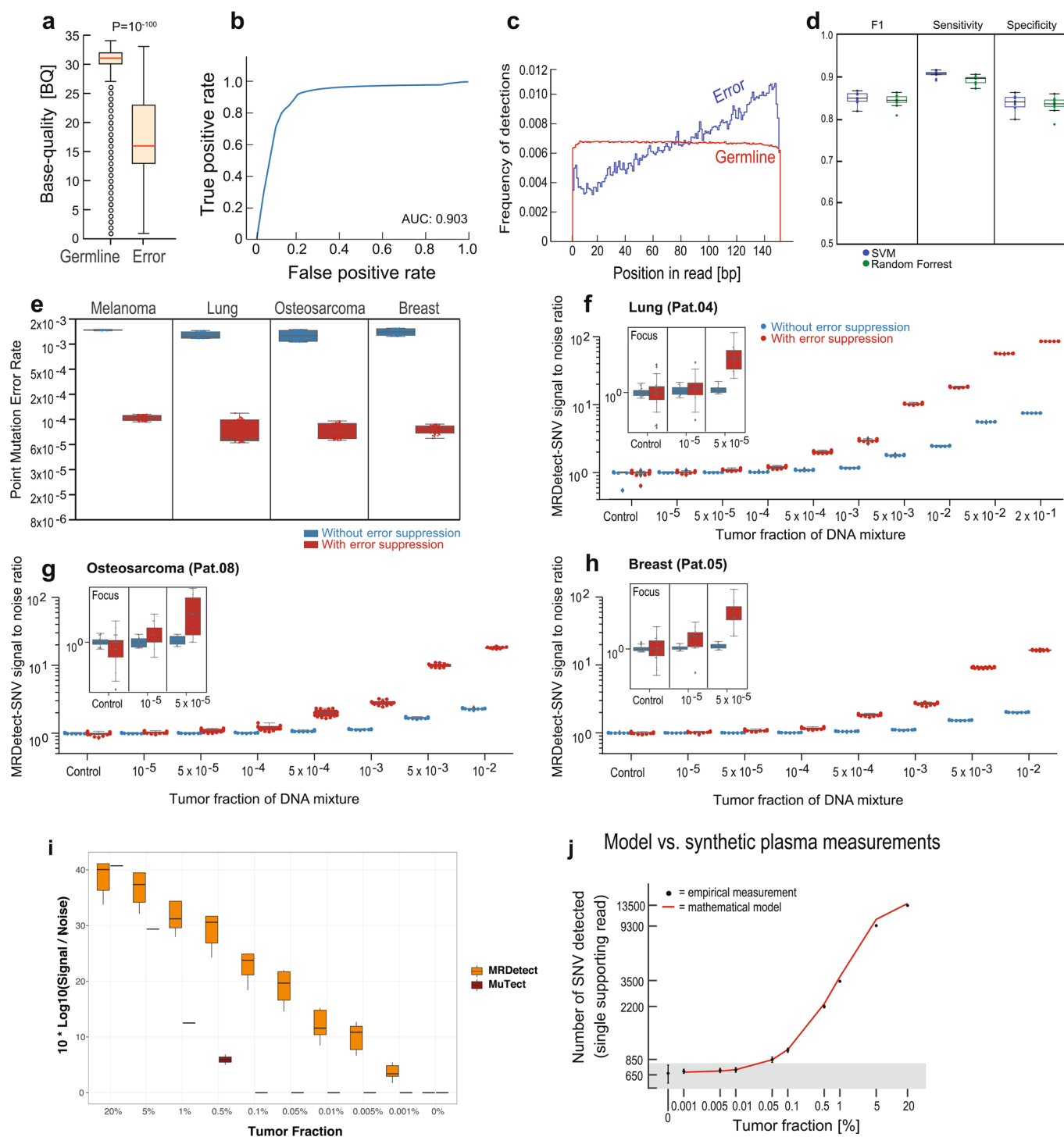
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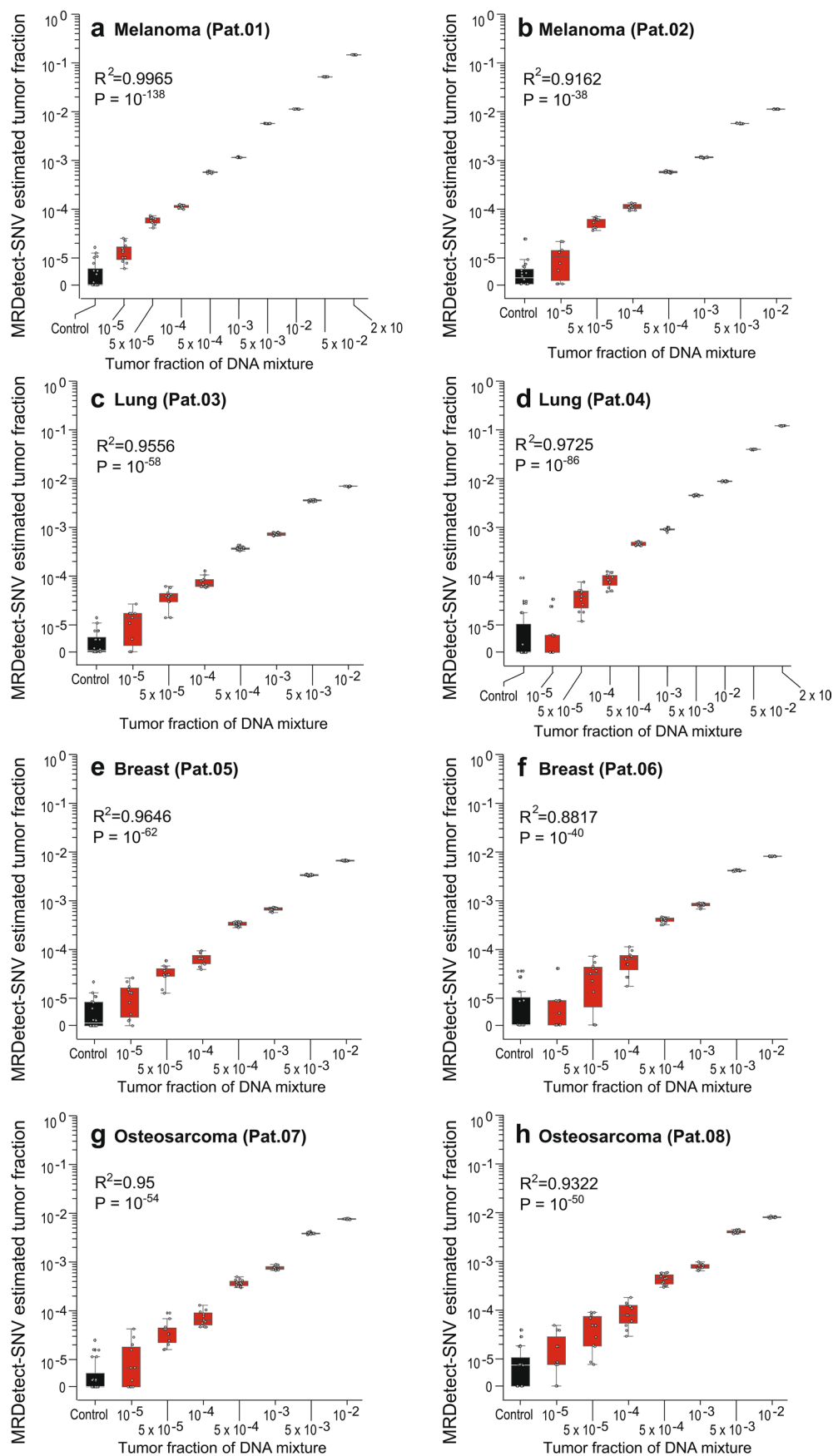


Extended Data Fig. 1 | Deep targeted sequencing analysis and DNA extraction optimization. **a-b**, constitute a re-analysis of previously published data¹⁸. **a**, Histogram of samples detected as a function of the maximum variant allele fraction (VAF) in stage I-IV cancer. Maximum VAF serves as an estimation of tumor fraction (TF). Patient histogram (solid bars) and empirical cumulative distribution (solid line) are shown, displaying stage dependent detection probability from 50% sensitivity in stage I cancer to >90% sensitivity in stage IV. Color coding represents different cancer stages. Number of patients considered per cancer stage is indicated in the figure. **b**, Histogram of variants (mutations) detected as a function of the variant allele fraction (VAF) for different cancer stages. Higher cancer stage exhibits an increase in the mutation VAF, associated with higher probability of detection. Number of mutations included in the analysis per cancer stage is indicated in the figure. **c**, Comparison of cfDNA yields across various leading commercial extraction kits. Multiple extractions (n=3) for each kit were performed over 1 mL aliquots of the same input plasma sample (from plasmapheresis of a normal donor), and total mass (per 1 mL) was determined with Qubit (ThermoFisher, Waltham, MA). Mean value and confidence interval (standard deviation) is shown. **d**, Extracted cfDNA was evaluated with Nanodrop (ThermoFisher, Waltham, MA) to detect impurities such as salt and genomic DNA. Omega Bio-Tek (Norcross, GA) had the lowest amount of contaminant carry over compared to other kits. Multiple tests (n=3, solid lines) for each kit were performed over 1 mL aliquots of the same input plasma sample.



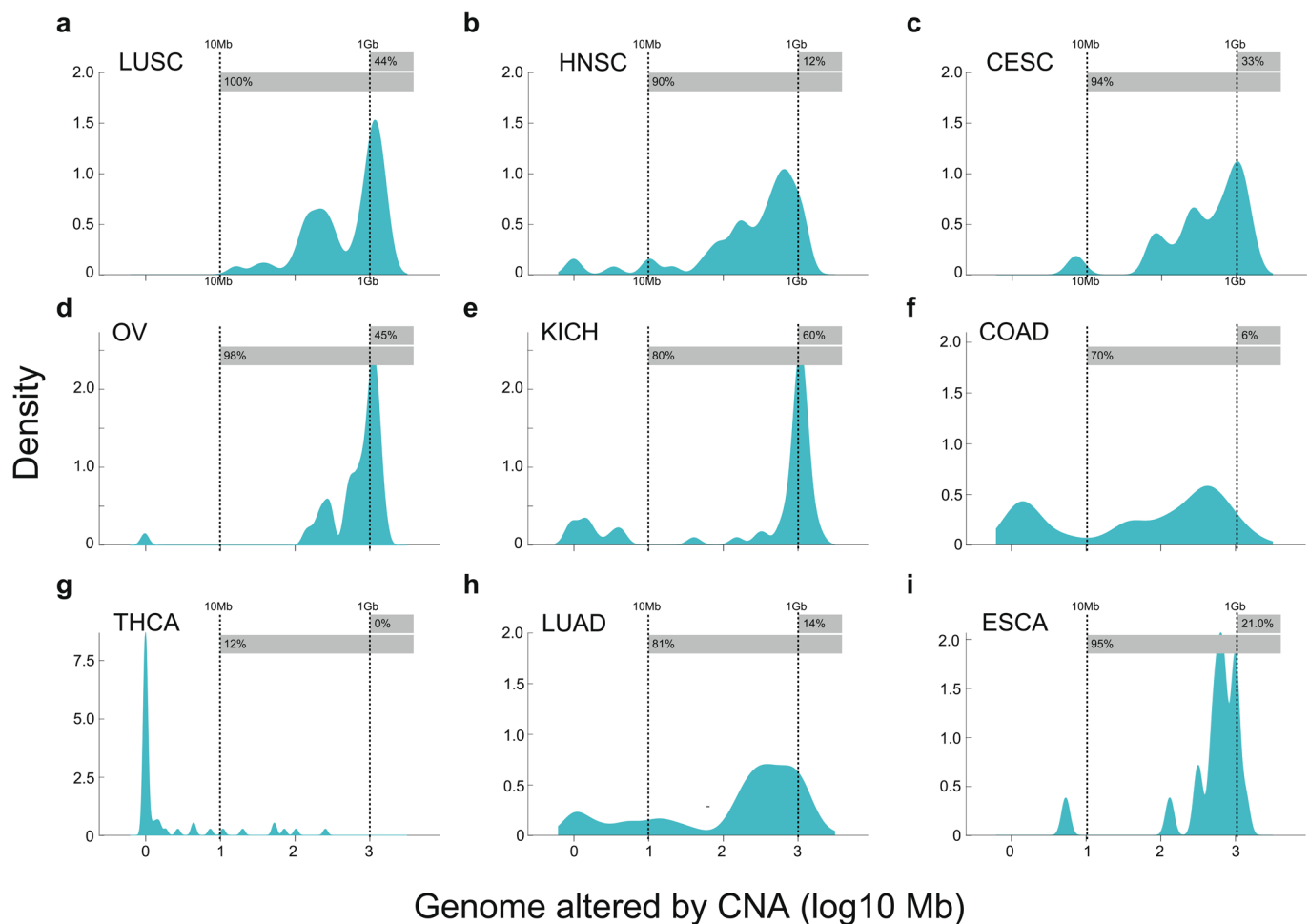
Extended Data Fig. 2 | See next page for caption.

Extended Data Fig. 2 | Patient-specific genome-wide SNV integration and error suppression. **a**, Base quality (BQ) signal for high quality germline single nucleotide polymorphisms (SNPs; $n = 9,142,326$) vs. low VAF (single supporting read) artifactual variants ($n = 407,061$) from representative PBMC sample (Pat.01), showing distinct separation between SNP and the low VAF artifacts BQ distributions (P value $< 10^{-100}$, two-sample t-test), and supporting effective filtration of sequencing artifacts by BQ filtration. **b**, Receiver-operating-curve (ROC) analysis for BQ filtering including germline SNPs (true labels) and low VAF artifactual variants (false labels), using the same data set as in **(a)**, and showing high filter performance for this simple quality metric ($AUC = 0.9$, $n = 9,142,326$). **c**, Variant position-in-read (PIR) shows association between low VAF artifactual variants ($n = 407,061$) and position at the 3' of the sequencing read, while germline SNPs ($n = 9,142,326$) show uniform spread across the sequencing read length. **d**, Support vector machine (SVM) classification performance between germline SNPs (random subsampling, $n = 100,000$) vs. low VAF artifactual variants (random subsampling, $n = 100,000$) from all PBMC WGS data ($n = 8$). Performance of SVM and random forest classification was compared over the same sample set with 10-fold cross validation. **e**, Box plot of error rate estimations before error suppression (blue) and after SVM-based error suppression (red) over four cancer types and 40 PBMC-derived replicates (2 patients per cancer type, 20 replicates per patient's PBMC sample). Error rate was calculated as the number of mismatches detected over the number of bp checked. Showing a uniform error reduction (median 14 fold-change reduction, range 11-17). **f-h**, Patient-specific SNV signal-to-noise quantification over a range of TFs (10^{-5} - 10^{-2}) compared to basal noise signal detected in control (TF = 0, subsampled PBMC DNA fragments) samples (left column). Signal-to-noise was estimated by calculating the log difference between the number of detections in each plasma-like admixture (TF > 0) and the mean number of detections in the controls (TF = 0). Analysis was done separately using tumor and matched germline (PBMC) WGS from lung (**f**, Pat.04), breast (**g**, Pat.05) and osteosarcoma (**h**, Pat.08) patients. Inset panel shows discrimination of tumor and control samples down to tumor fraction 10^{-5} after utilizing machine-learning-based sequencing error suppression (red) vs. reduced sensitivity with the raw unfiltered data (blue). **i**, Benchmarking of mutation detection performance for mutation centric method²³ vs. read-centric method (MRDetect). Patient-specific SNV signal-to-noise quantification over a range of TFs (10^{-5} - 2×10^{-1}) compared to basal noise signal detected in control (TF = 0, subsampled PBMC DNA fragments) samples (right column). Signal-to-noise was estimated by calculating the log difference between the number of detections in each plasma-like admixture (TF > 0) and the mean number of detections in the controls (TF = 0). **j**, Single nucleotide variant (SNV) point mutation detection in plasma mixtures with different tumor fractions (TF > 0) and controls (TF = 0) is shown. Y-axis shows the number of detections (variants observed in tumor WGS and also detected in plasma synthetic admixture) as a function of TF (x-axis). Red line constitutes the number of detections predicted for each TF based on the mutation load, coverage, noise model. Gray area represents the area under the background noise model threshold (1.5std), showing robust discrimination from noise for TF > 10^{-3} . Analysis was done on 35X coverage lung cancer (Pat.04) admixture cohort. Centre values represent mean and error bars represent standard deviation. In (f-j), $n = 11$ independent admixture samples for TFs > 0 and $n = 20$ independently down-sampled PBMC replicates for the control (TF = 0) of each patient. Throughout the figure, boxplots represent median, bottom and upper quartile; whiskers correspond to 1.5 x IQR.

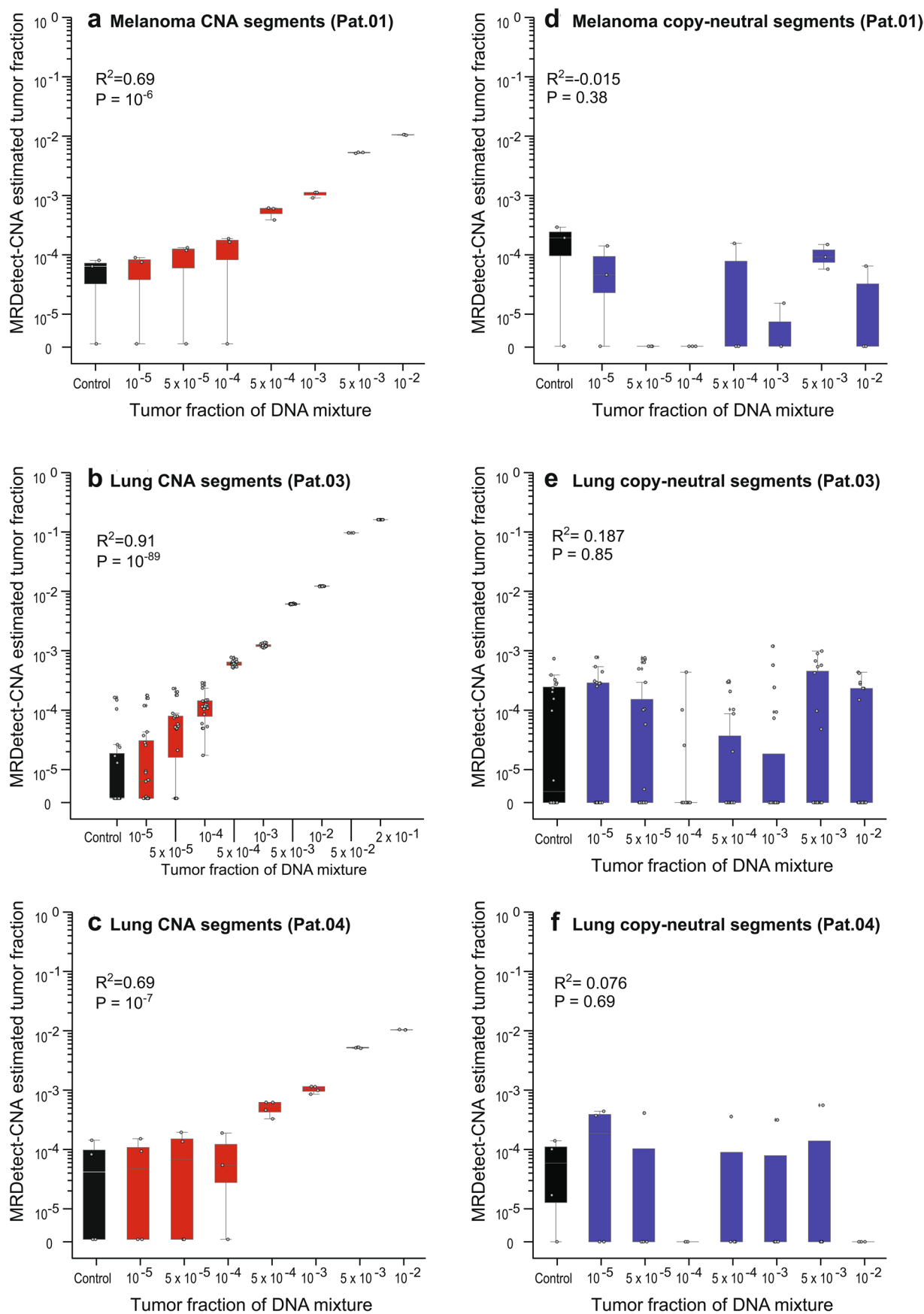


Extended Data Fig. 3 | See next page for caption.

Extended Data Fig. 3 | Patient-specific genome-wide SNV integration provides accurate read out of tumor fraction. a-h, Tumor fraction (TF) inference plots using MRDetect genome-wide SNV integration for two melanomas (**a**: Pat.01, **b**: Pat.02), two lung cancers (**c**: Pat.03, **d**: Pat.04), two breast cancers (**e**: Pat.05, **f**: Pat.06), and two osteosarcomas (**g**: Pat.07, **h**: Pat.08). Each plot was generated using *in silico* admixtures of varying TF (range 10^{-5} -0.2) with 35X depth of coverage by randomly downsampling and mixing tumor reads (mean coverage 97x, range 85x-110x) and germline reads (mean coverage 49x, range 43x-56x) from WGS data (see Methods; Supplementary Table 1). For TFs > 0, n=11 independent admixture samples. For the control (TF=0), n=20 independently down-sampled PBMC replicates. We observed accurate TF estimation as low as 10^{-5} , discriminated from control (TF=0) samples (left box-plot), with high Pearson correlation (two-sided test) between the input TF mixture (x-axis) and the SNV-based estimated TF prediction, confirming accurate inference based on genome-wide mutational integration. Throughout the figure, boxplots represent median, bottom and upper quartile; whiskers correspond to 1.5 x IQR.

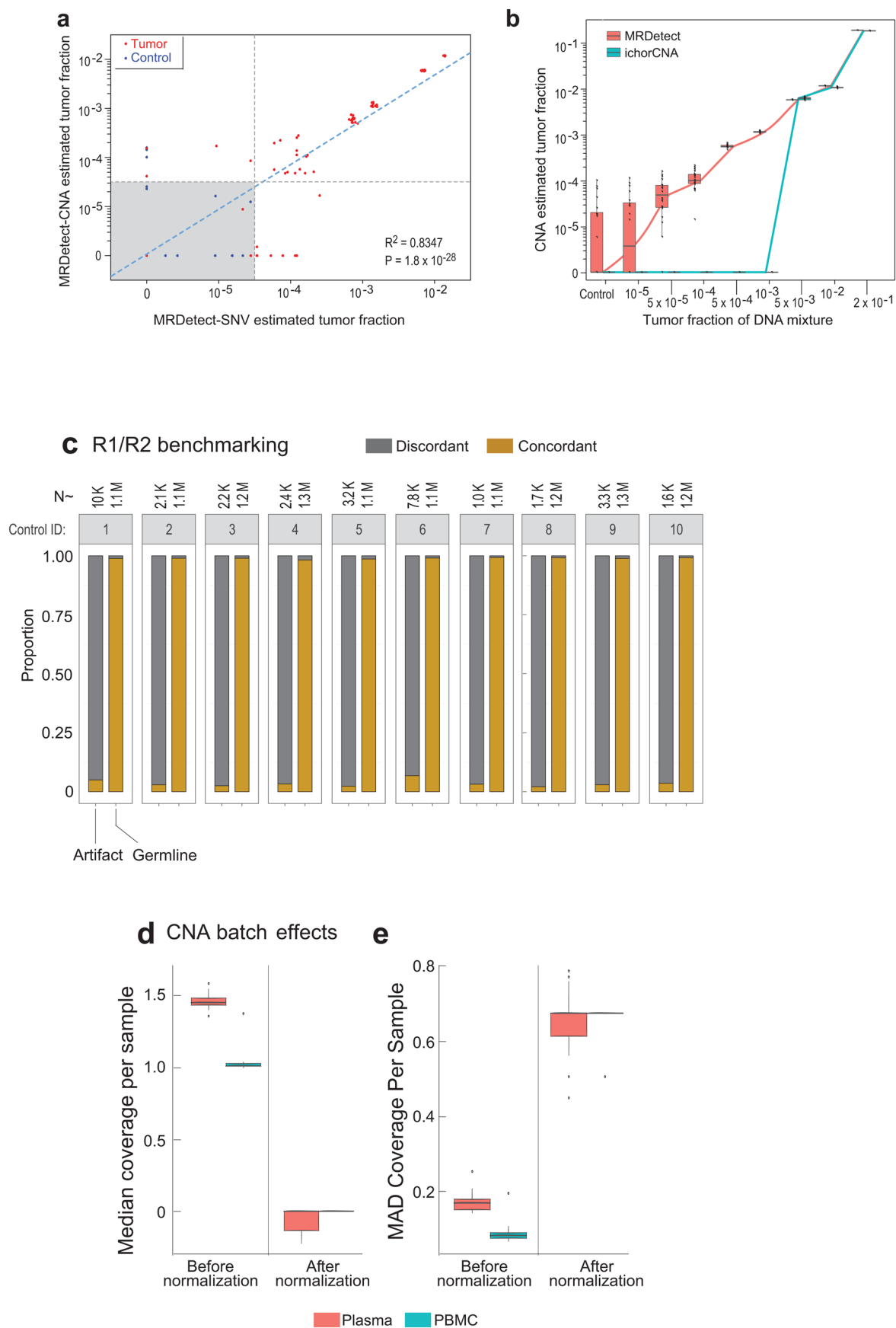


Extended Data Fig. 4 | CNA load across tumor types. Histogram of CNA load in WGS samples across cancer types from the TCGA cohort. Measured as a function of the size of genome altered by CNA (in $\log_{10}$ Mb). Dashed lines represent the percentage of samples that have CNA load of over 10 Mb and 1 Gb respectively, for each cancer type. Cancer types include LUSC: Lung squamous cell carcinoma ($n=50$), HNSC: Head and Neck squamous cell carcinoma ($n=50$), CESC: Cervical squamous cell carcinoma and endocervical adenocarcinoma ($n=18$), OV: Ovarian serous cystadenocarcinoma ($n=50$), KICH: Kidney Chromophobe ($n=50$), COAD: Colon adenocarcinoma ($n=53$), THCA: Thyroid carcinoma ($n=50$), LUAD: Lung adenocarcinoma ($N=152$), ESCA: Esophageal carcinoma ($n=19$).



Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | CNA based detection of ctDNA tumor fraction. a-c, Tumor fraction (TF) inference using MRDetect genome-wide CNA integration for representative patients, including melanoma (**a**: Pat.01) and lung cancer (**b**: Pat.03, **c**: Pat.04). Each plot was generated using *in silico* admixtures of varying TF (range 10^{-5} – 0.2) in 18X coverage, by randomly downsampling and mixing tumor reads (mean coverage 97X, range 85X–110X) and germline reads (mean coverage 49X, range 43X–56X) from WGS data (see Methods; Supplementary Table 1). Twenty replicates used for each TF > 0 sample, and for the control (TF = 0) samples, showing accurate TF estimation as low as 5×10^{-5} , discriminated from control (TF = 0) samples (left box-plot), with high Pearson correlation between the input TF mixture (x-axis) and the CNA-based estimated TF prediction (two-sided test). **d-f,** Tumor fraction (TF) inference in neutral regions (no copy number gain or loss in the tumor WGS data) for the same *in silico* admixtures (**d**: melanoma, Pat.01; **e**: lung, Pat.03; **f**: lung, Pat.04), shows the expected low Pearson correlation between input admixture TF and the signal (two-sided test), consistent with no expected coverage changes in the plasma admixtures in these regions. Throughout the figure, boxplots represent median, bottom and upper quartile; whiskers correspond to $1.5 \times$ IQR.

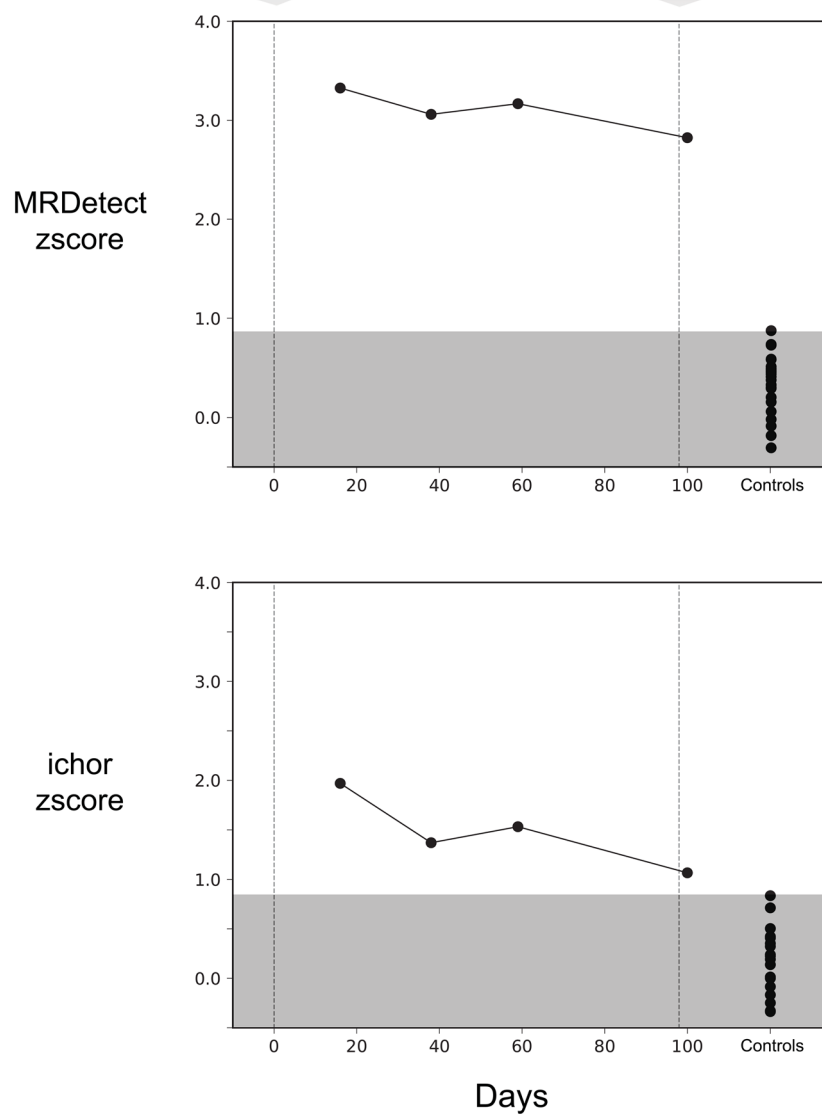
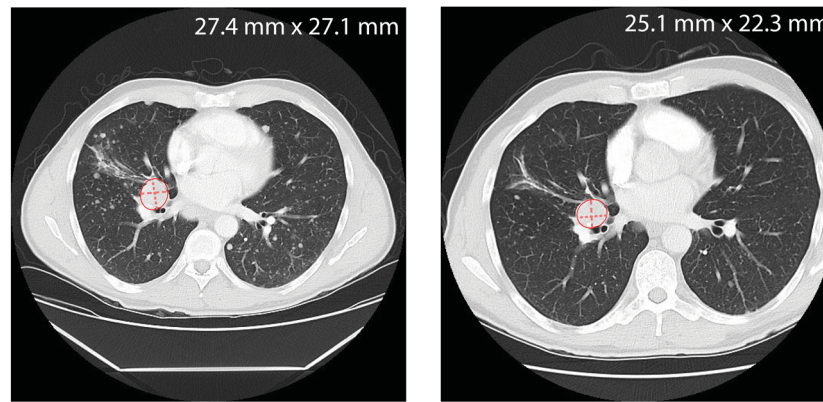


Extended Data Fig. 6 | See next page for caption.

Extended Data Fig. 6 | CNA and SNV MRDetect correlation and further error suppression. **a**, Spearman correlation between SNV and CNA TF estimation across TF admixtures for a lung tumor (Pat.03) shows high correlation (two-sided test) between the two orthogonal inference methods. Red dots correspond to cancer plasma (TF > 0) samples and blue dots correspond to control plasma (TF = 0) samples. Detection threshold (dashed lines) were set on $TF < 5 \times 10^{-5}$ for both methods. Eleven replicates used for each TF > 0 sample, and 20 replicates were used for the control (TF = 0) samples. **b**, Comparison to an orthogonal CNA-based TF method- ichor-CNA¹⁵. Analyzing the same cohort of breast cancer (Pat.05) in silico synthetic admixture samples shows concordance in TF estimation for high TF (TF > 5×10^{-3}), with extension of detection for MRDetect to lower TFs. The same 20 replicates were used for both MRDetect and ichor-CNA for each of the TF > 0 and control (TF = 0) samples. **c**, Proportion of variant concordant (brown) vs. discordant (gray) read pairs (R1 and R2) detected in germline SNPs and artifactual variants. Analysis was done across 10 control (benign lung lesions) plasma samples, comparing read-pairs associated with germline SNPs (right bar) vs. read-pairs associated with artifactual variants (left bar) per plasma sample. The artifactual variants were defined by read pairs with variants overlapping the union of all patient somatic SNVs compendia across all LUAD patients, that were observed with the same variant in the control plasma sample. The number of read-pairs used in the analysis is indicated above each bar. **d**, Median genome-wide normalize (divided by mean coverage) coverage from matched germline PBMC WGS samples from patients with LUAD (cyan, n = 15) and control plasma WGS samples from patients with benign lung lesions (red, n = 11) before and after robust Z-score normalization. **e**, Median absolute deviation (MAD) calculated over normalized-coverage (*that is*, divided by mean coverage) from matched germline PBMC and control plasma WGS samples as in (d) before and after robust Z-score normalization. Throughout the figure, boxplots represent median, bottom and upper quartile; whiskers correspond to $1.5 \times IQR$.

a

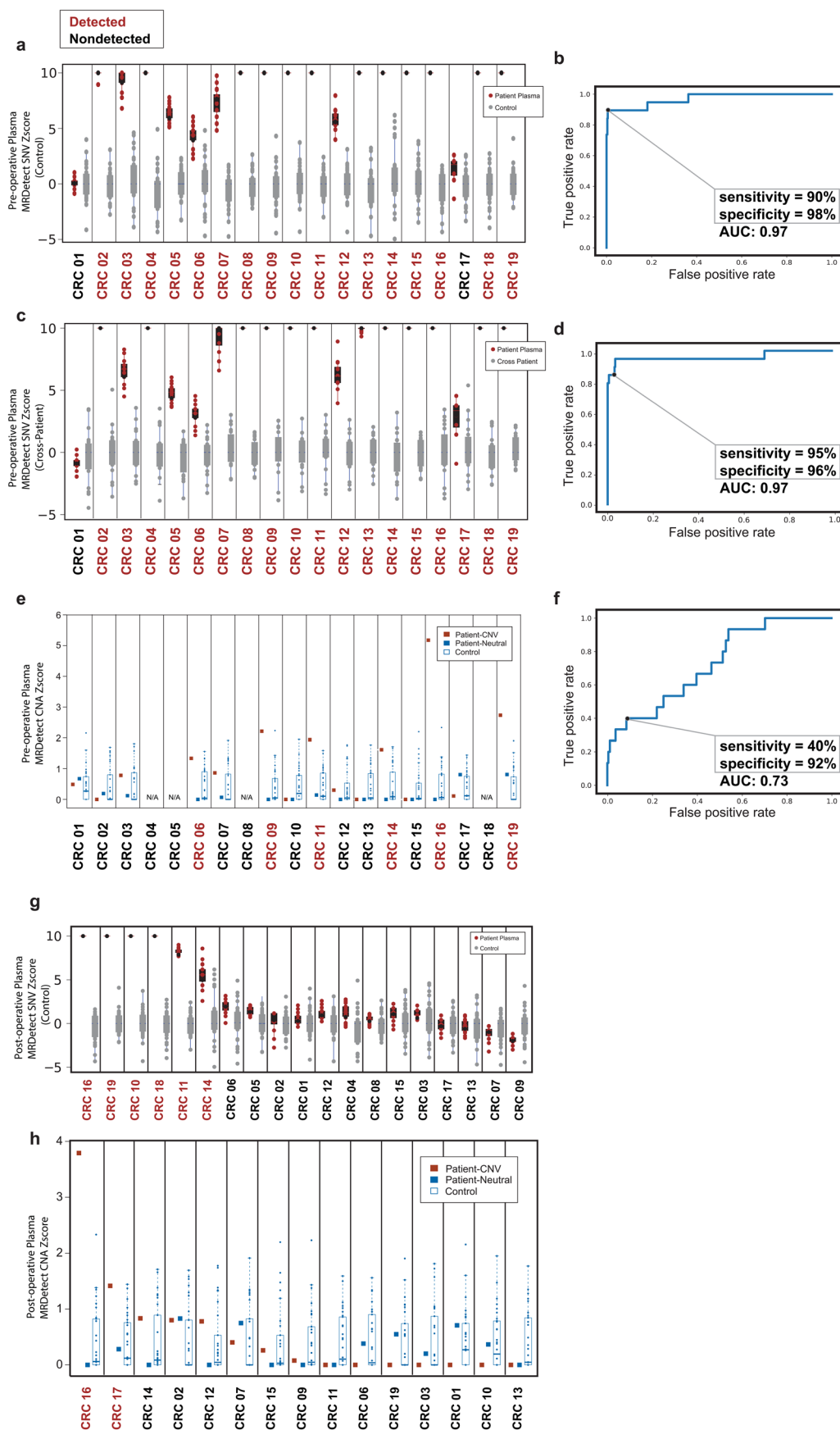
MEL02



Nivolumab/Ipilimumab

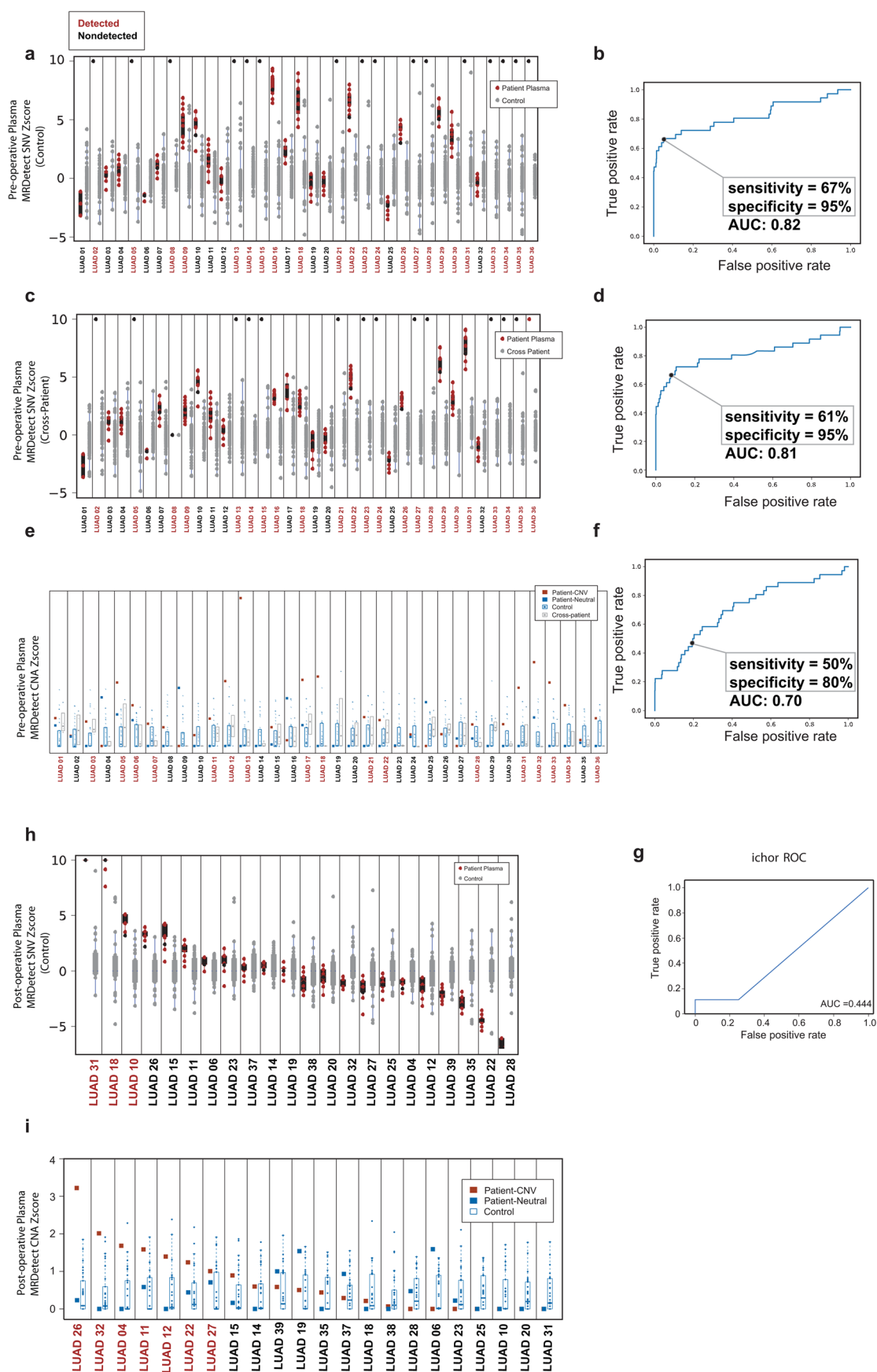
Extended Data Fig. 7 | See next page for caption.

Extended Data Fig. 7 | Serial application of MRDetect to monitor melanoma response to immunotherapy. Melanoma treatment response (Patient MELO2) during immunotherapy (Nivolumab) is monitored by blood samples. Upper panels- Treatment monitoring by computed tomography (CT) shows response to therapy but residual disease after 3 months of therapy. Middle panel- MRDetect Z-scores effectively track tumor responses, matching radiographic changes, in higher temporal resolution than that feasible with imaging. Lower panel- ichor-CNA captures treatment response dynamics but showing lower signal to noise ratio compared to the MRDetect method. For both MRDetect and ichor-CNA methods, log Z-score is calculated from a single plasma sample for each timepoint compared to a panel of control samples ($n=30$). Throughout the figure, boxplots represent median, bottom and upper quartile; whiskers correspond to $1.5 \times$ IQR.



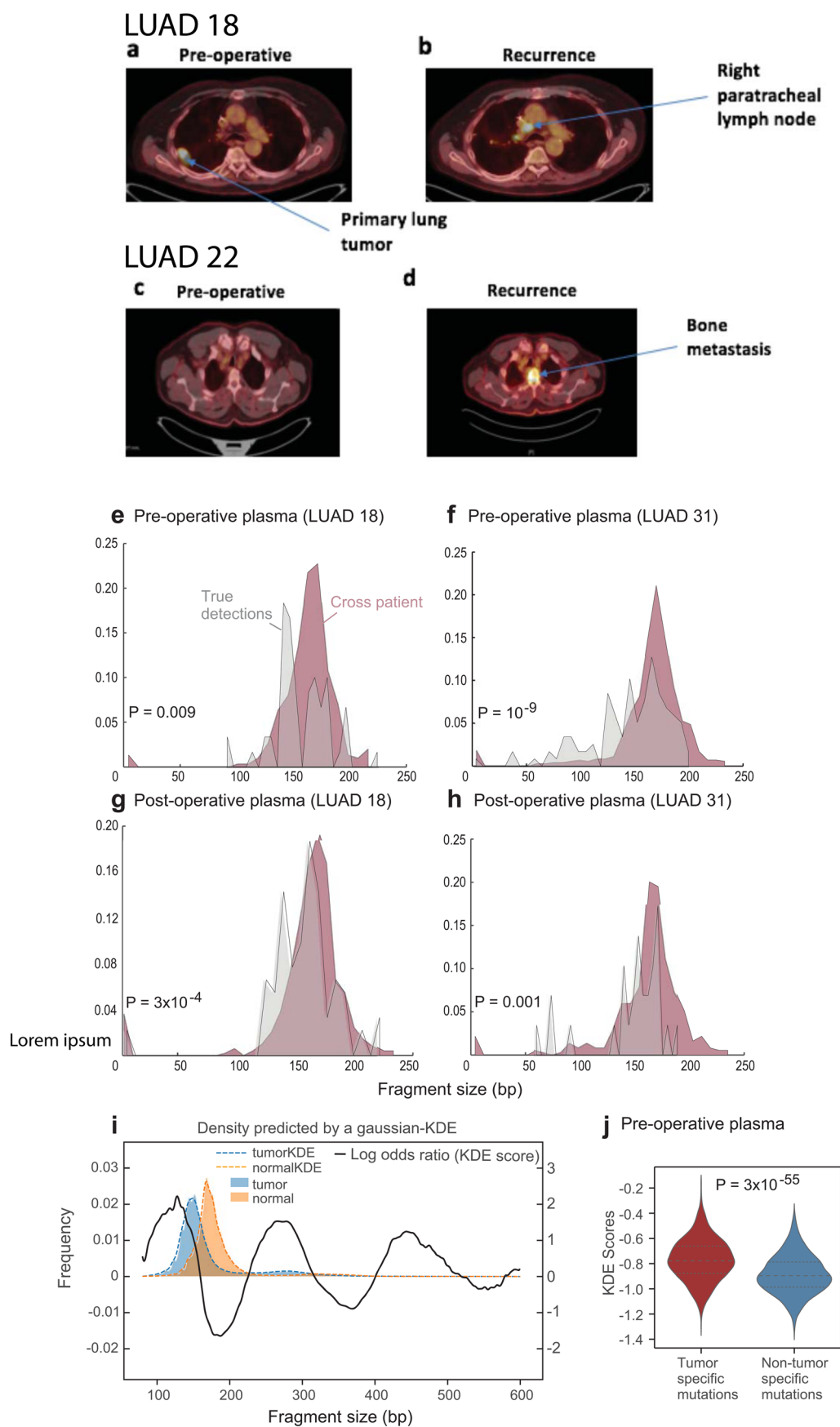
Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | MRDetect performance in colorectal monitoring. **a**, Robust Z-score discrimination between signal detected across 20 random subsamplings (80% of reads per subsampling iteration) of LUAD patient pre-operative plasma (black, $n=36$ patients) and the cohort of control plasma test set (gray, $n=30$). The signal was measured on the subsampling set and control test set using the same patient-specific point mutation (SNV) compendium. Z-score was calculated using the noise parameters estimated in the control test cohort (see Methods). **b**, Receiver-operating-curve (ROC) analysis was performed over all SNV-based Z-score values calculated on the patients' pre-operative plasma and control plasma as in **(a)**. **c**, Cross patient noise evaluation. Robust Z-score discrimination between signal detected at 20 random subsamplings (80% of reads per subsampling iteration) of LUAD patient pre-operative plasma WGS (black, $n=36$ patients), cross-patient noise estimation via application of the patient-specific compendium to all other patient pre-operative plasma ($n=35$, gray). Z-score was calculated using the noise parameters estimated in the cross-patient cohort (see Methods). **d**, Receiver-operating-curve (ROC) analysis was performed over all SNV-based Z-score values calculated on the matched patients and cross-patient plasma. **e**, Z-score discrimination between MRDetect-CNA on LUAD patient pre-operative plasma (red, $n=36$ patients) compared to signal detected in neutral regions (as a negative control, blue), control plasma test cohort ($n=30$) and cross-patient cohort ($n=35$). Cross-patient noise was estimated by applying the patient-specific CNA compendium to other patient plasma samples ($n=35$, all other patients). Z-score was calculated using the noise parameters estimated by the control plasma cohort. **f**, Receiver-operating-curve (ROC) analysis was performed over all CNA-based Z-score values calculated on the patients' pre-operative plasma and control patients. **g**, ROC analysis was performed over all ichor-CNA⁴ TF values calculated on the LUAD patients' pre-operative plasma and control patients ($n=66$). Interestingly the two patient plasma samples detected by ichor-CNA included events that do not appear in the tumor, one of them was found to be a PBMC specific somatic event (potentially from clonal hematopoiesis). Throughout the figure, boxplots represent median, bottom and upper quartile; whiskers correspond to $1.5 \times$ IQR. **h**, Z-score discrimination between signal detected in 20 random subsampling (80% of reads per subsampling iteration) of LUAD patient plasma WGS ($n=22$ patients) collected at a median of 17 days after surgery and a cohort of control plasma test samples (gray, $n=30$). The signal was measured on the matched plasma and control set using the same patient-specific point mutation (SNV) compendium. Z-score was calculated using the noise parameters estimated in the control cohort (see Methods). **i**, Z-score discrimination between MRDetect-CNA on LUAD patient plasma (red, $n=22$ patients) collected at a median of 17 days after surgery, compared to signal detected in neutral regions (as a negative control, blue) and control plasma cohort ($n=30$). Z-score was calculated using the noise parameters estimated by the control plasma cohort (see Methods). Throughout the figure, boxplots represent median, bottom and upper quartile; whiskers correspond to $1.5 \times$ IQR.



Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | MRDetect performance in LUAD monitoring. **a**, Robust Z-score discrimination between signal detected across 20 random subsamplings (80% of reads per subsampling iteration) of LUAD patient pre-operative plasma (black, $n = 36$ patients) and the cohort of control plasma test set (gray, $n = 30$). The signal was measured on the subsampling set and control test set using the same patient-specific point mutation (SNV) compendium. Z-score was calculated using the noise parameters estimated in the control test cohort (see Methods). **b**, Receiver-operating-curve (ROC) analysis was performed over all SNV-based Z-score values calculated on the patients' pre-operative plasma and control plasma as in **(a)**. **c**, Cross patient noise evaluation. Robust Z-score discrimination between signal detected at 20 random subsamplings (80% of reads per subsampling iteration) of LUAD patient pre-operative plasma WGS (black, $n = 36$ patients), cross-patient noise estimation via application of the patient-specific compendium to all other patient pre-operative plasma ($n = 35$, gray). Z-score was calculated using the noise parameters estimated in the cross-patient cohort (see Methods). **d**, Receiver-operating-curve (ROC) analysis was performed over all SNV-based Z-score values calculated on the matched patients and cross-patient plasma. **e**, Z-score discrimination between MRDetect-CNA on LUAD patient pre-operative plasma (red, $n = 36$ patients) compared to signal detected in neutral regions (as a negative control, blue), control plasma test cohort ($n = 30$) and cross-patient cohort ($n = 35$). Cross-patient noise was estimated by applying the patient-specific CNA compendium to other patient plasma samples ($n = 35$, all other patients). Z-score was calculated using the noise parameters estimated by the control plasma cohort. **f**, Receiver-operating-curve (ROC) analysis was performed over all CNA-based Z-score values calculated on the patients' pre-operative plasma and control patients. **g**, ROC analysis was performed over all ichor-CNA⁴ TF values calculated on the LUAD patients' pre-operative plasma and control patients ($n = 66$). Interestingly the two patient plasma samples detected by ichor-CNA included events that do not appear in the tumor, one of them was found to be a PBMC specific somatic event (potentially from clonal hematopoiesis). Throughout the figure, boxplots represent median, bottom and upper quartile; whiskers correspond to $1.5 \times$ IQR. **h**, Z-score discrimination between signal detected in 20 random subsampling (80% of reads per subsampling iteration) of LUAD patient plasma WGS ($n = 22$ patients) collected at a median of 17 days after surgery and a cohort of control plasma test samples (gray, $n = 30$). The signal was measured on the matched plasma and control set using the same patient-specific point mutation (SNV) compendium. Z-score was calculated using the noise parameters estimated in the control cohort (see Methods). **i**, Z-score discrimination between MRDetect-CNA on LUAD patient plasma (red, $n = 22$ patients) collected at a median of 17 days after surgery, compared to signal detected in neutral regions (as a negative control, blue) and control plasma cohort ($n = 30$). Z-score was calculated using the noise parameters estimated by the control plasma cohort (see Methods). Throughout the figure, boxplots represent median, bottom and upper quartile; whiskers correspond to $1.5 \times$ IQR.



Extended Data Fig. 10 | See next page for caption.

Extended Data Fig. 10 | Imaging of sample monitored LUAD cases and fragment length analysis. Positron emission tomography-computed tomography (PET-CT) of two patients (LUAD#3 and LUAD#6) confirms no radiographically observable metastatic spread at the time of surgery (**a** and **c**, respectively), while metastatic recurrence has been identified approximately six months post-operative by PET-CT (**b** and **d**, respectively). **e-h**, Representative fragment size histograms showing the distribution of DNA fragments as a function of the fragment length. DNA fragments that are associated with tumor mutations (gray) are showing significantly shorter size compared to DNA fragments that are associated with artifactual non patient-specific detections (red, derived from applying cross-patient mutational compendia; median P value $< 10^{-3}$, two-sample t-test). Plots include two pre-operative plasma samples (**e**: LUAD#18, tumor mutation detection $n=35$, artifactual detection $n=18802$; **f**: LUAD#31, tumor mutation detection $n=82$, artifactual detection $n=32530$) and matching plasma samples after surgery (**g**: LUAD#18, tumor mutation detection $n=55$, artifactual detection $n=22737$; **h**: LUAD#31, tumor mutation detection $n=27$, artifactual detection $n=14610$). **i**, Kernel-density-estimator (KDE) trained to discriminate between tumor-derived (human aligned reads from a patient derived xenograft model, see Methods, blue) and normal-derived (from control plasma samples, orange) cfDNA based on the fragment size signature. The log difference between the tumor and normal density functions (black solid line) was used as a score function that integrates the fragment size shift signal across the entire fragment sizes distribution (80bp-600bp). **j**, Pre-operative cfDNA showed significant shift (two-sample t-test) in their tumor-specific mutation detections in the patient plasma (red, $n=563$), compared to non-tumor (cross-patient) detected mutations in the same samples (blue, $n=4184$). Violin plots depict kernel density estimates of the density distribution. Center dashed lines represent the median and dashed lines represent the interquartile range.

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Software and code

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Data collection

All samples were collected at WCM, MSKCC, or MGH, and were sequenced at NYGC.

Data analysis

Data was analyzed using Illumina's bcl2fastq (v2.17.1.14), BWA-ALN (v0.6.2), GATK (v3.4.0), Samtools (v1.1), MuTect (v1.1.7), Strelka (v1.0.14), LoFreq (v2.1.3a), NBIC-seq (v0.7), dbSNP (build 151), Ichor-CNA (v2.0), Conpair (v0.2), JaBbA (v0.0.0.9), Picard (v1.83), Novosort MarkDuplicates (v1.03.01)
Custom python scripts (python version 3.6.2) using sklearn's SVM package, numpy.metrics.roc_curve, numpy.metrics.roc_auc_score, scipy.stats.gaussian_kde.
Custom R script (R version 3.4.2) using fitdistrplus and lm function
Custom MRDetect software is available as described in Code Availability Statement.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes used in our study are sufficient to assess differences between cancer plasma cfDNA and control plasma cfDNA , giving us enough statistical power to detect differences in patient-specific mutational signature detection rates. In silico admixture were analyzed with 20 replicates for each condition, and patient signature detection included 96 cancer patient samples against 38 control plasma samples (split into training and validation sets of 8 and 30 , respectively). We estimate that this cohort is powered to detect an effect size of >30% increase in hazard ratio between the post-operative detected and not detected patient groups.
Data exclusions	To specifically assess potential sample contamination we applied Conpair, which validated genetic concordance between matched germline, tumor and plasma samples to confirm that all samples came from the same individual, as well as evaluates potential inter-individual contamination in the samples. Samples that showed low concordance (<.99) were excluded from further analysis. Specifically, three pre-operative plasma samples from LUAD patients #37, #38, and #39 were rejected from analysis due to low concordance score. This exclusion criterion was pre-established.
Replication	In silico admixture were analyzed with 20 replicates for each condition. Patient specific signature detection was checked over every cancer patient sample against 30 control plasma samples. Error suppression was trained with a training set of 8 control plasma samples, and evaluated against a held-out test set of 30 control plasma samples. Error suppression showed consistent performance on independent biological replicates.
Randomization	Randomization was not included in this retrospective, non-intervention study.
Blinding	Blinded analysis was not included in this study to allow for ROC based optimization of detection thresholds. However, to safeguard against potential biases, we performed cross-patient analysis and observed consistent error rates and detection performance without any experimental group bias.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

Methods

n/a	Involved in the study	n/a	Involved in the study
☒	☐ Antibodies	☒	☐ ChIP-seq
☒	☐ Eukaryotic cell lines	☒	☐ Flow cytometry
☒	☐ Palaeontology	☒	☐ MRI-based neuroimaging
☐	☒ Animals and other organisms		
☐	☒ Human research participants		
☒	☐ Clinical data		

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	PDX were established using fresh pathological tissue fragments from a patient with triple negative breast cancer, implanted subcutaneously into six- to eight-week-old anesthetized NGS female mice.
Wild animals	This study did not involve wild animals.
Field-collected samples	This study did not involve samples collected from the field.
Ethics oversight	Mouse PDX studies were reviewed and approved by Institutional Animal Care and Use Committee (IACUC) at Weill Cornell Medicine.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Human research participants

Policy information about [studies involving human research participants](#)

Population characteristics	Control cohort (n = 38) had a median age of 67 and was 42% female. Lung adenocarcinoma cohort (n = 39) had a median age of 73 and was 53% female. Colorectal cohort (n = 19) had a median age of 60 and was 32% female. All patients in both Lung and Colorectal cohorts underwent surgical tumor resection. Melanoma cohort had two patients, both male, aged 37 and 58, and both underwent systemic therapy.
Recruitment	Lung adenocarcinoma and control patient samples were collected from the NewYork-Presbyterian/Weill Cornell Medical Center Thoracic surgery department. Melanoma patient samples were collected from Memorial Sloan Kettering Cancer Center. Colorectal Cancer patient samples were collected at Massachusetts General Hospital. The study was offered to all individuals undergoing surgery at collection centers prior to surgery and we are not aware of any self-selection bias.
Ethics oversight	The study was approved by the local ethics committee (WCM, MGH and MSKCC) and by the Institutional Review Boards (IRB) and conducted in accordance to the Declaration of Helsinki protocol.

Note that full information on the approval of the study protocol must also be provided in the manuscript.